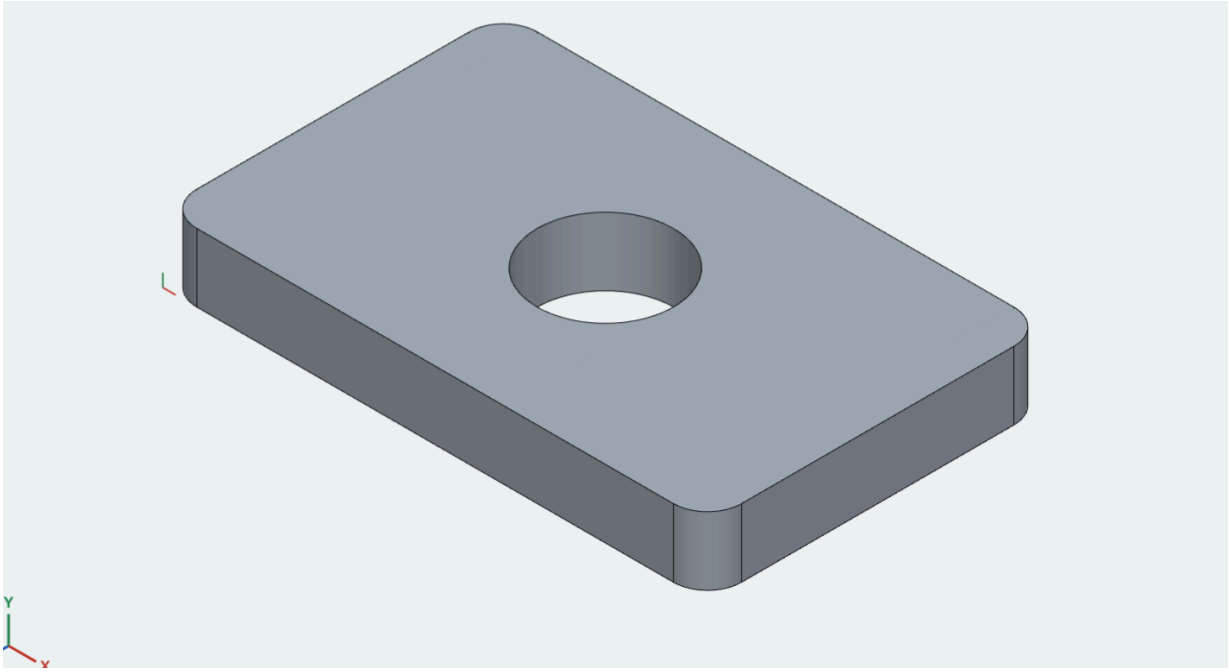
Parametric 3D modeler for schools

Banco CAD

User manual



Sketches · Features · Assemblies · Drawings

Updated on 12 November 2026 · © 2026 Rayuk-creator · rayuk-creator.github.io/CAD

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1 What Banco CAD is

Banco CAD is a parametric 3D modeler designed for schools, in particular for technical schools of mechanics and mechatronics. It takes after SolidWorks in everything: command names, way of working, feature tree, panels, sketch colors and drawings to ISO standards. What you learn here, you find again in professional programs.

“Parametric” means that the part is built as a history of steps you can edit: change a sketch dimension or the depth of an extrusion, and the whole part is recalculated.

1.1 What you can do

- **Draw sketches** with lines, rectangles, circles, arcs and polygons, constrained with dimensions and relations.
- **Build parts** with extrusions, cuts, revolves, sweeps, ribs, standard holes, fillets, chamfers, patterns and mirrors, also with several solid bodies.
- **Evaluate** mass, volume and center of mass with 13 materials, measure and section the part.
- **Build assemblies** with mates, subassemblies, standard screws and bolts, bill of materials, interference detection and exploded view.
- **Make drawings** with the European method (ISO 5456-2, first angle projection), with dimensions, sections and title block, and save them as PDF, SVG or DXF.
- **Exchange files** with other programs: STEP, IGES, STL, OBJ, DXF and DWG can be opened; STEP and STL can be written.

1.2 How to open it

The program is **a single file**, with nothing to install. There are two ways to use it.

Way	What to do	When it is best
From the website	Open **rayuk-creator.github.io/CAD** in the browser. For the English version add **?lang=en** at the end of the address, or choose **Project › Language: English**.	It updates by itself; it also works on tablets and phones.
From the file	On github.com/Rayuk-creator/CAD open the file index.html, press **Download raw file**, rename it for example banco-cad.html and open it with a double click.	It works without the Internet: for example at school, in a lab with no network.

> **Watch out** The right address is **rayuk-creator.github.io/CAD**. On the repository page (github.com/Rayuk-creator/CAD) the program does not run, and it looks half loaded.

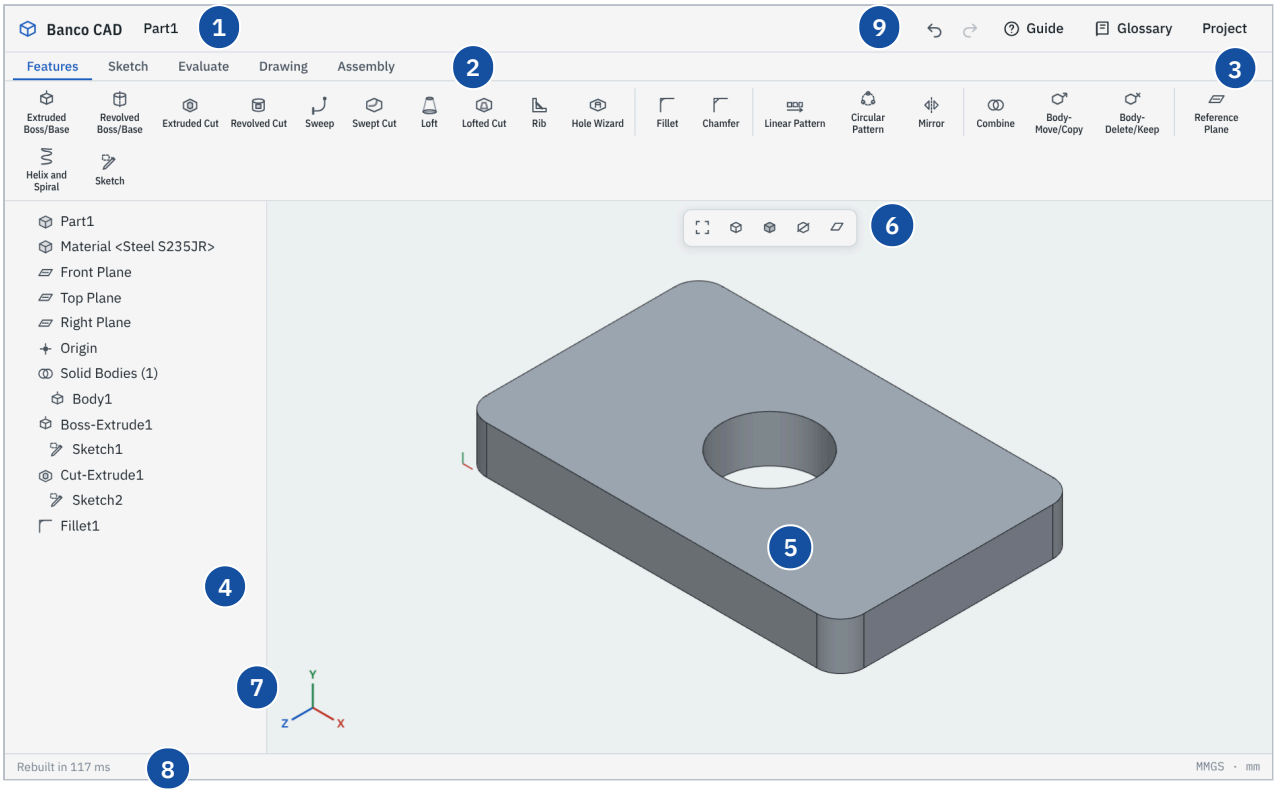
You need a recent browser with 3D graphics (WebGL) enabled. The automatic tests of the program run on Chromium, the engine of Chrome and Edge.

1.3 Where your work goes

Your work is saved **by itself in the browser** you are using, on that computer. All the computing happens there: the program asks nothing of other websites and your parts stay on your computer. Three practical rules follow from this.

- Work done on the website and work done with the downloaded file **stay separate**, even on the same computer. The same goes from one browser to another.
- To take a project to another computer, or to keep it safe, use **Project › Export project**; to open it again, **Project › Import a project**.
- If you clear the browser data, you can lose your work. Export the project now and then: it is your backup copy.

2 The screen and the commands



The parts of the screen. The numbers match the table below.

	Part	What it is for
1	**Name and open part**	The name of the part. With several parts or assemblies, the list to switch from one to another appears here.
2	**Tabs**	Features, Sketch, Evaluate, Drawing, Assembly: each one shows its commands.
3	**CommandManager**	The command buttons of the chosen tab.
4	**FeatureManager design tree**	Planes, origin, bodies and features in the order you made them. When you create or edit a feature, the **PropertyManager** appears here, that is the panel with the parameters.
5	**3D view**	The part. Select with a click; with a panel open, a small tree also appears over the view, to choose by name.
6	**View toolbar**	Zoom to fit, orientations, display style, section, planes.
7	**Axes**	They show how the part is turned: X red, Y green, Z blue.
8	**Status bar**	It says what is selected and, in a sketch, whether it is defined.
9	**Undo, Guide, Glossary, Project**	Undo and Redo; the “how to” Guide; the dictionary of terms; the menu with saving, importing and exporting.

In the panels ✓ confirms and ✕ cancels. The small ⓘ next to a title explains that item.

2.1 Moving the view

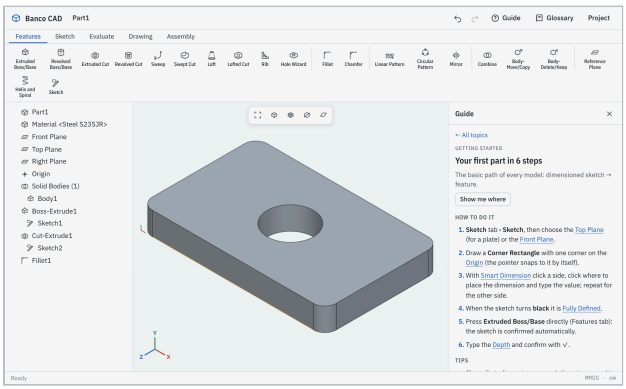
Mouse	Action
Wheel forward / back	Zooms in / out towards the pointer (also in the drawing)
Ctrl + wheel	Rotates the part
Drag with the left button	Rotates the part
Drag with the right button, or Shift + left	Pans the view
Click	Selects; Ctrl + click selects more than one

Tablet	Action
Two fingers, always	Zoom
Touch the part first (“Part held”, blue border), then one finger	Rotate
Touch the part first, then two fingers	Pan
Tap on empty space or “Release”	Releases the part

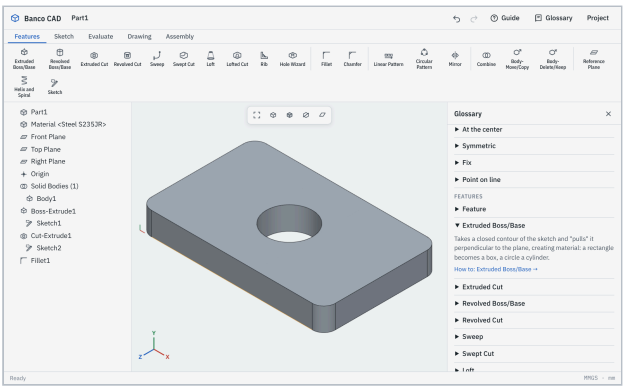
Keyboard	Action
F	Fits the part to the window
Spacebar	Opens the view orientations
Ctrl + Z / Ctrl + Y	Undo / Redo (also Ctrl + Shift + Z)
Esc	Cancels the command or closes the panel; in a sketch goes back to selection
Enter	Confirms the open panel; in a sketch confirms the sketch
Delete	Deletes what is selected

2.2 Guide and Glossary inside the program

This manual uses the texts of the **Guide** and of the **Glossary**, which are always at hand in the program. The Guide has the **Show me where** button, which points to the command on the screen. The Glossary explains every term; from the panel titles it opens with the ⓘ.



The Guide stays open on the right while you work.



The Glossary, with the link to “how to”.

2.3 All the commands, tab by tab

Sketch tab

Command	What it does
Line	Consecutive lines: click for each vertex, double-click or Esc to finish
Corner Rectangle	Rectangle given by two opposite corners
Center Rectangle	Rectangle given by the center and a corner
Circle	Circle: center, then a point on the circumference
Centerpoint Arc	Arc: center, start point, end point
3 Point Arc	Arc: start point, end point, point on the arc
Tangent Arc	Tangent arc starting from the end of a line or an arc
Polygon	Inscribed regular polygon: center, then a vertex
Straight Slot	Straight slot: center of the first arc, center of the second, width
Point	Sketch point
Centerline	Dash-dot construction line: axis of revolution or of symmetry
Sketch Fillet	Rounds the corner between two lines: click the vertex
Sketch Chamfer	Bevels the corner between two lines: click the vertex
Trim Entities	Click the part of the entity to remove (up to the intersections)
Mirror Entities	Select the entities and a centerline, then press here
Convert Entities	Copies the selected edges of the face (in gray) into the sketch
Insert DXF or DWG	Brings lines, arcs, circles, ellipses and splines of a DXF or DWG file into the sketch
Construction Geometry	Turns the selected entities into construction geometry (and back)
Smart Dimension	Click one or two entities, then click to place the dimension and type the value
Add Relation	Select the entities and choose the relation in the panel on the left
Exit Sketch	Confirms the sketch (like the ✓ at the top right)

Features tab

Command	What it does
Extruded Boss/Base	Extrudes a closed sketch and adds material
Revolved Boss/Base	Rotates a profile around a centerline
Extruded Cut	Removes material by extruding a sketch
Revolved Cut	Removes material by rotating a profile
Sweep	Moves a closed profile along a path (tangent lines and arcs): tubes, bent springs, handles
Swept Cut	Removes material by moving a profile along a path: channels, grooves
Loft	Joins two or more closed profiles on different planes: twisted blades, transitions between different sections, hoppers
Lofted Cut	Removes material by passing through two or more closed profiles on different planes

Command	What it does
Rib	Thin reinforcing wall from an open line to the part
Hole Wizard	Simple holes, clearance holes for screws, counterbores, countersinks and tapped holes (ISO)
Fillet	Rounds the selected edges
Chamfer	Bevels the selected edges
Linear Pattern	Repeats features along one or two directions
Circular Pattern	Repeats features around an axis
Mirror	Copies features symmetrically about a plane
Combine	Joins two or more bodies, subtracts one from another, or keeps only the common part
Body-Move/Copy	Moves, rotates or copies a solid body
Body-Delete/Keep	Removes bodies, or keeps only the chosen ones
Reference Plane	Plane parallel to a plane or a face, at a given distance
Helix and Spiral	Helix or spiral curve starting from a circle: it is the path of springs and threads
Sketch	New sketch on the selected plane or face

Evaluate tab

Command	What it does
Mass Properties	Volume, area, mass and center of mass
Measure	Distances, diameters and angles between the selected entities
Material	Chooses the material of the part
Section View	Cuts the view with a plane
Self-check	Rebuilds the test parts and compares volume, area, center of mass and bounding box with the values calculated by hand (1% tolerance)

Drawing tab

Command	What it does
Dimension	ISO dimensions: click an edge, a circle or two points, then click to place
Overall dimensions	Adds the overall dimensions of the views and the diameters of the circles
Section View	Replaces a view with its section (cutting plane and hatching)
Fit the sheet	The whole sheet in the window
Save PDF	PDF file of the drawing, vector and to scale (real sheet, A4–A2): it opens and prints anywhere
Export SVG	Vector drawing at 1:1 scale on paper
Export DXF	For AutoCAD and other 2D CAD programs (ISO layers and line types)
Print / PDF	Prints the drawing to scale (or saves to PDF from the browser)

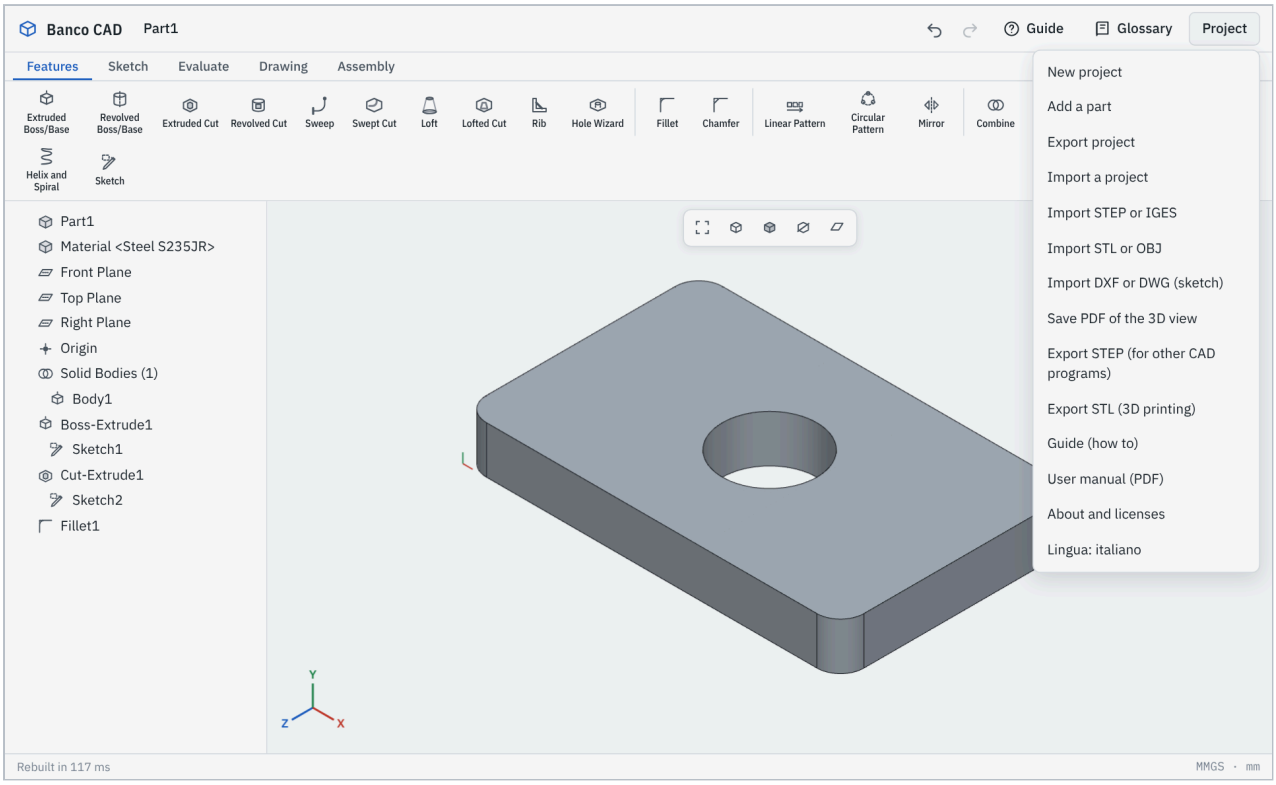
Assembly tab

Command	What it does
Insert Part	Adds a part of the project or a file (STEP, IGES, STL, OBJ, project) as a component of the assembly
New Part	Creates an empty part in this project and opens it
New Assembly	Creates another assembly in this project; it can become a subassembly of another one
Screws and Bolts	Inserts a standard screw, nut or washer, with a cosmetic thread
Screw in Hole	One click on the hole: inserts the right size screw and mates it automatically
Mate	Links two entities of two components: coincident, concentric, parallel, perpendicular, distance, angle, tangent
Move Component	Moves or rotates a component in its remaining degrees of freedom (also on tablets)
Mass Properties	Mass, volume, area and center of mass of the whole assembly and of each component
Bill of Materials	The list of the parts of the assembly, with quantity, material and mass
Interference Detection	Finds the components that overlap and by what volume
Exploded View	Moves the components apart to show how they fit together, without changing their real positions
Measure	Distances, diameters and angles between the selected entities
Section View	Cuts the view with a plane

Project menu

Item	What it does
New project	Starts an empty project.
Add a part	Adds another part to the same project.
Delete this part	Removes the open part and its components in the assemblies. It cannot be undone.
Export project	Saves the project to a file: it is the backup copy and the way to move it.
Import a project	Opens a saved project again.
Import STEP or IGES	Opens a solid from a STEP or IGES file.
Import STL or OBJ	Opens a solid made of triangles from an STL or OBJ file.
Import DXF or DWG (sketch)	Brings a 2D drawing into a sketch.
Save PDF	Saves the drawing, the sketch or the 3D view as PDF, depending on what you are looking at.
Export STEP (for other CAD programs)	Writes the part to open it in another CAD program.
Export STL (3D printing)	Writes the part for 3D printing.
Guide (how to)	Opens the Guide.
User manual (PDF)	Opens this manual as PDF (an Internet connection is needed).
About and licenses	Shows authors and licenses.

Item	What it does
Lingua: italiano	Switches to Italian; the page reloads.



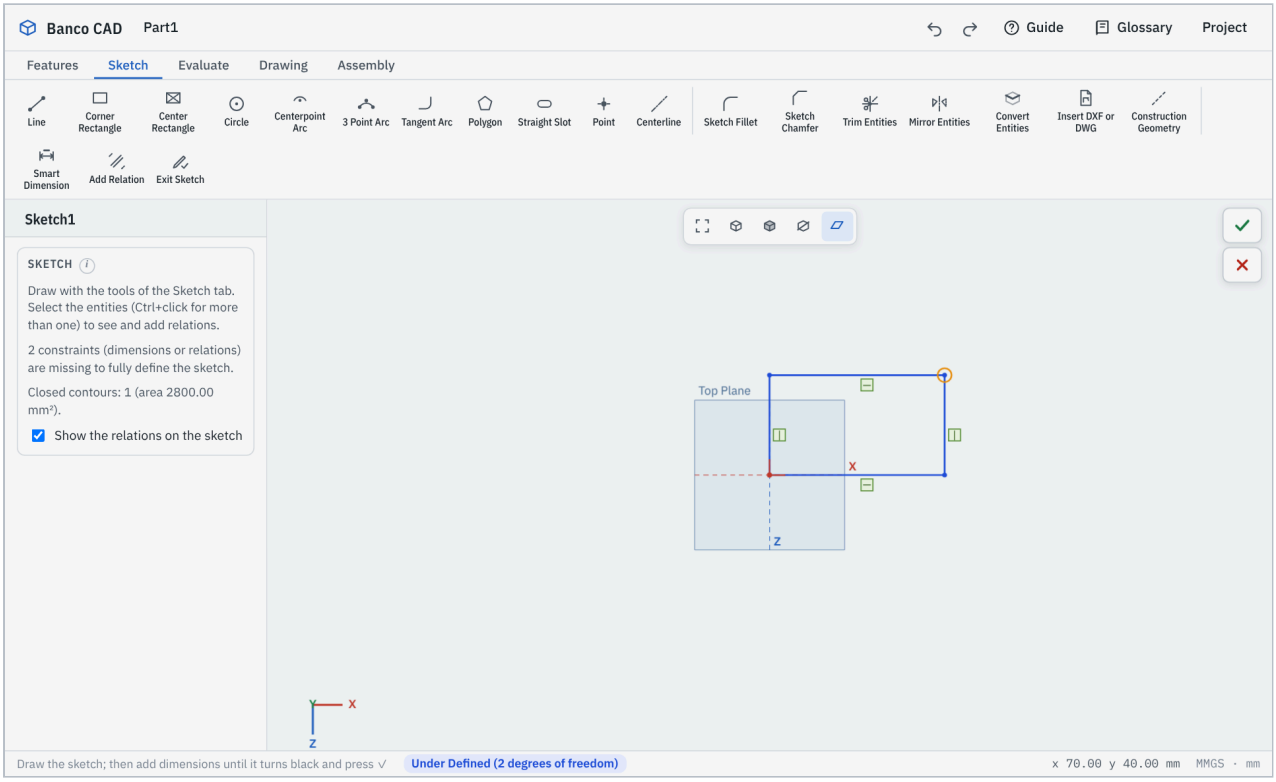
The Project menu opens with the button at the top right. The name of “Save PDF” changes depending on what you are looking at.

3 Guided exercise: a plate with a hole

The fastest way to learn is to make a real part from start to finish. We will build a plate **80 × 50 × 10 mm** with a central hole **Ø 20** and the four vertical edges filleted **R 5**. At the end we will check the numbers with a hand calculation. All the steps were really carried out in the program, with the mouse.

3.1 Draw the rectangle

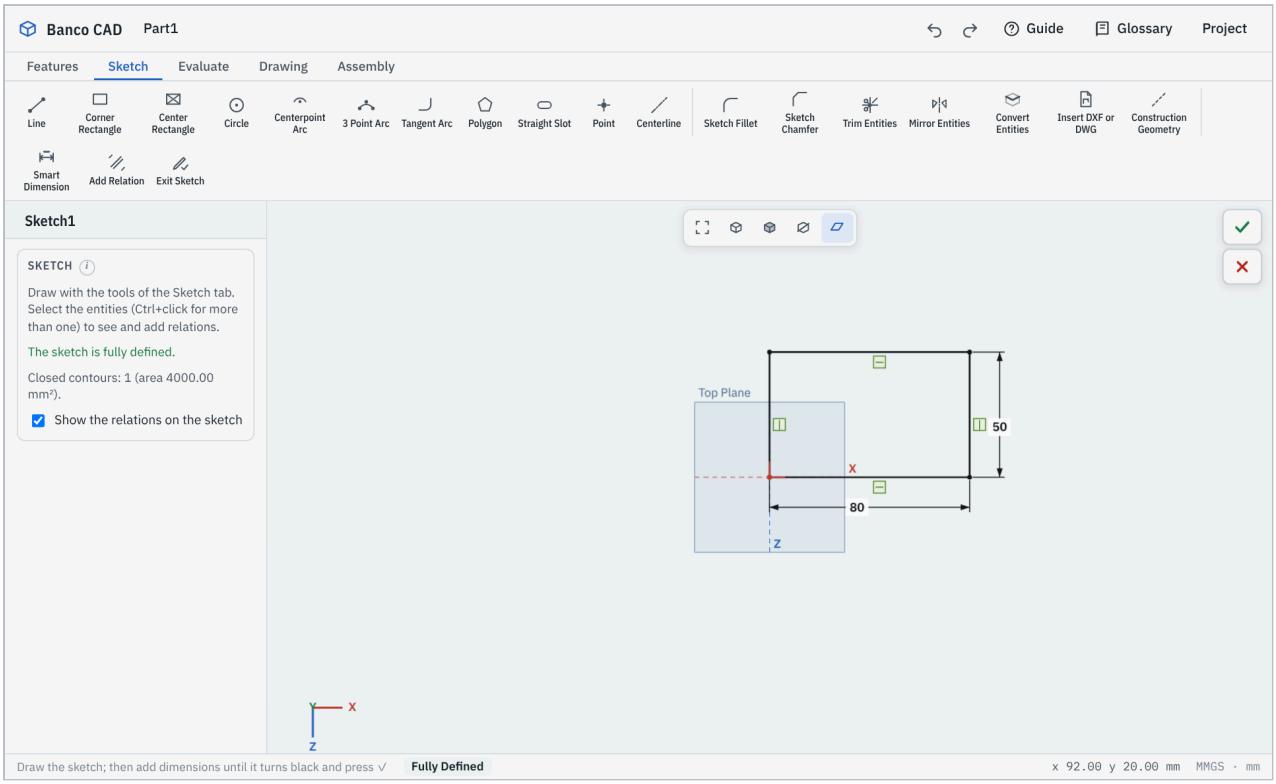
1. Open the **Sketch** tab, click **Top Plane** in the tree, then press **Sketch**. The view turns by itself.
2. Press **Corner Rectangle**. Click the origin (the pointer snaps to it by itself), then click a point up and to the right. The sizes do not matter: the dimensions will set them.
3. Press **Esc** to go back to selection. The lines are **blue**: the sketch is not defined yet.



Rectangle just drawn: blue means “it can still move”.

3.2 Add the dimensions

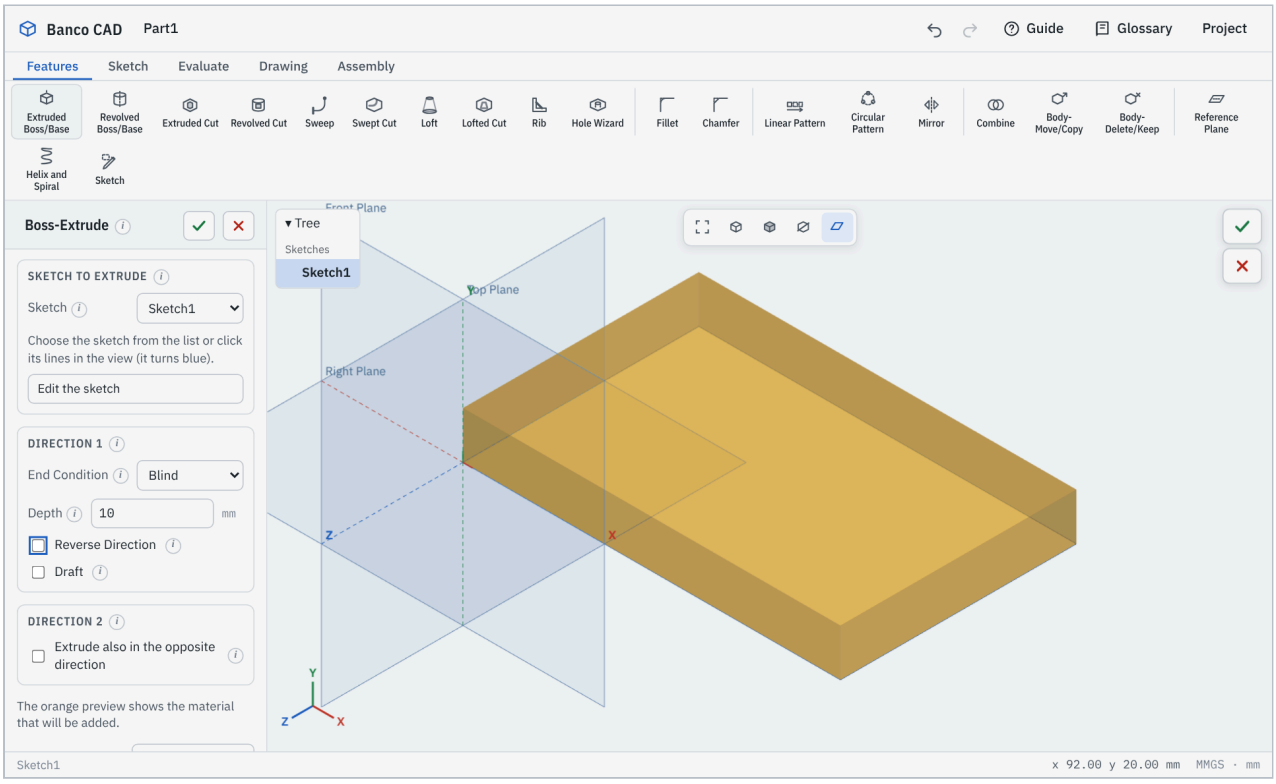
1. Press **Smart Dimension**. Click the bottom side, then below the side to place the dimension. Type **80** and press **Enter**.
2. Click the right side, then to its right. Type **50** and press **Enter**.
3. Now the lines are **black** and at the bottom it says **Fully Defined**.



Black sketch: shape and position are completely set.

3.3 Extrude

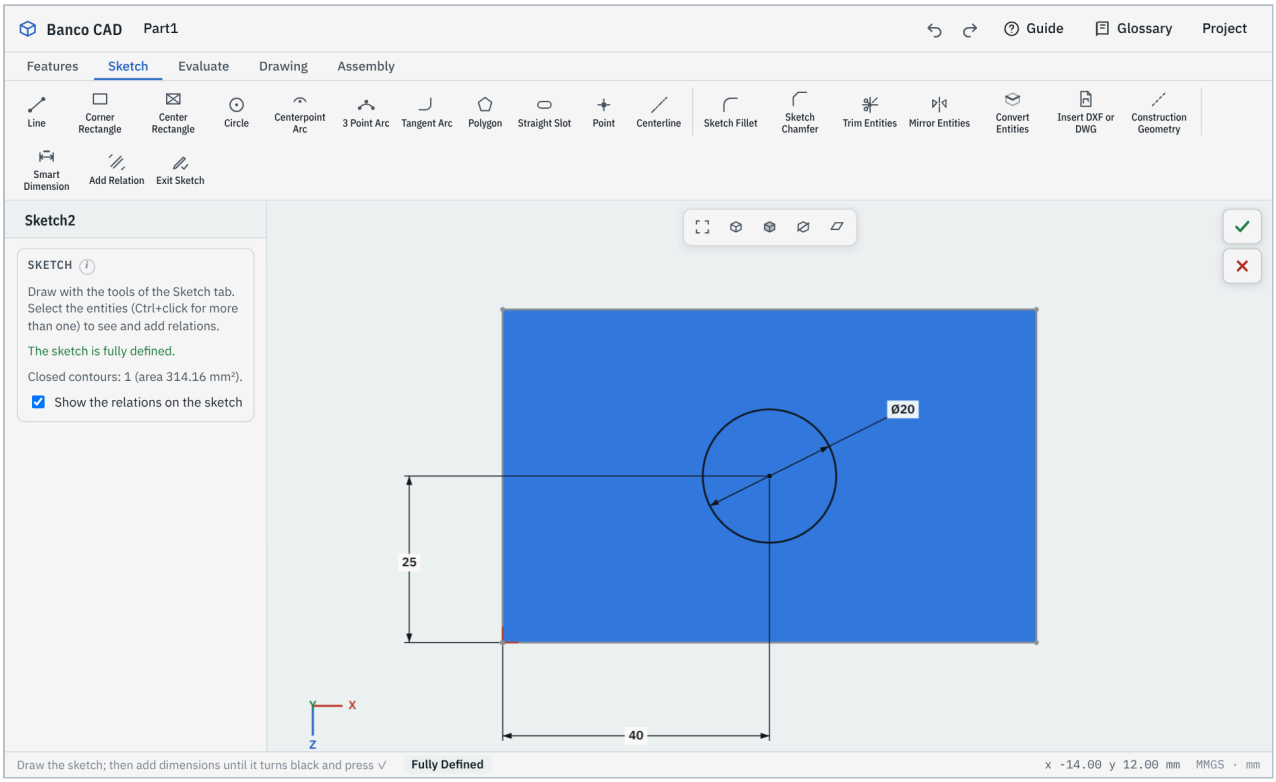
1. Open the **Features** tab and press **Extruded Boss/Base**: the sketch is confirmed by itself.
2. In the panel type **10** in **Depth**. The orange preview shows the material that will be added.
3. Confirm with **✓**. The part is born, and in the tree there is **Boss-Extrude1** with its sketch inside.



The extrusion panel with the preview.

3.4 Draw the hole on a face

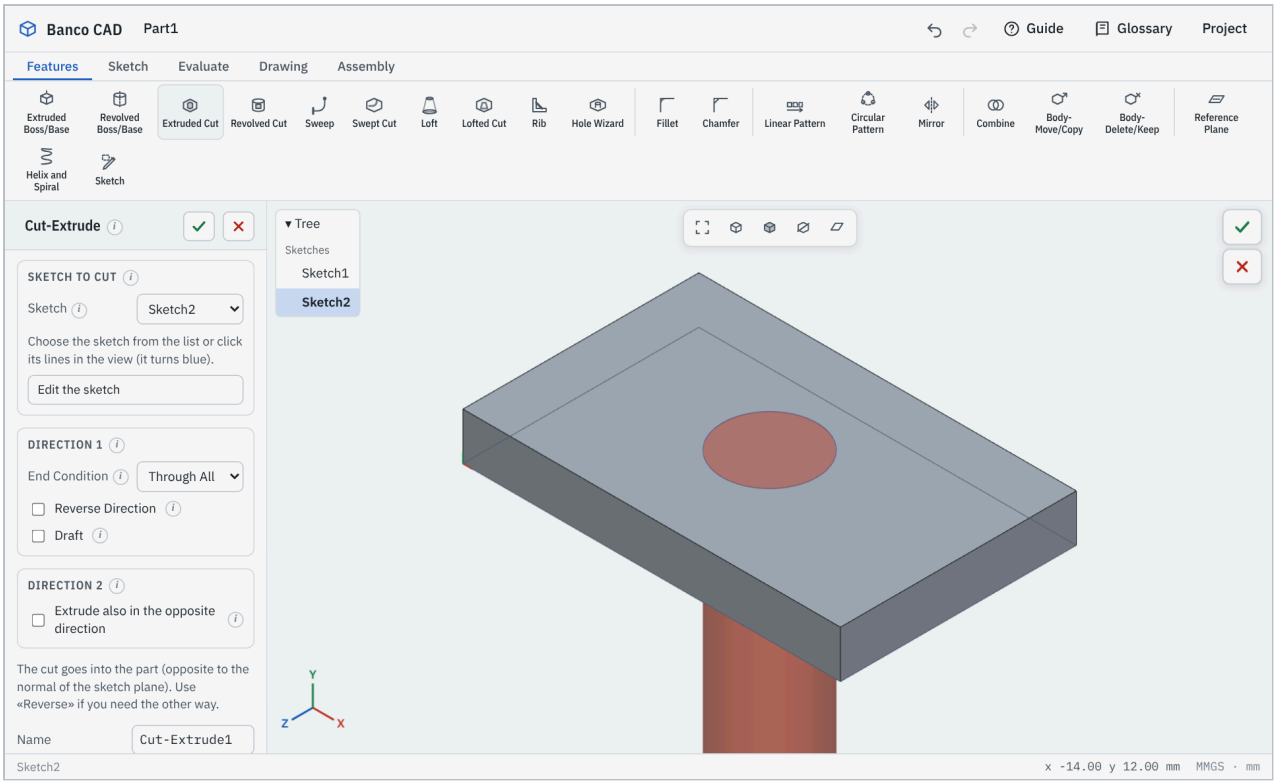
1. Click the top face of the plate. In the **Sketch** tab press **Sketch**: the new sketch is made on that face.
2. Press **Circle**. Click a point near the center, then a point a few millimeters away: that is the radius. Press **Esc**.
3. Press **Smart Dimension**. Click the circle, then outside it, and type **20**.
4. Click the origin, then the center of the circle, then below the plate to place the horizontal dimension: type **40**.
5. Again: click the origin, click the center, then to the left of the plate for the vertical dimension: type **25**. The sketch turns black.



Circle Ø 20 with the two dimensions that put it in the center (40 and 25). The chosen face stays light blue.

3.5 Cut

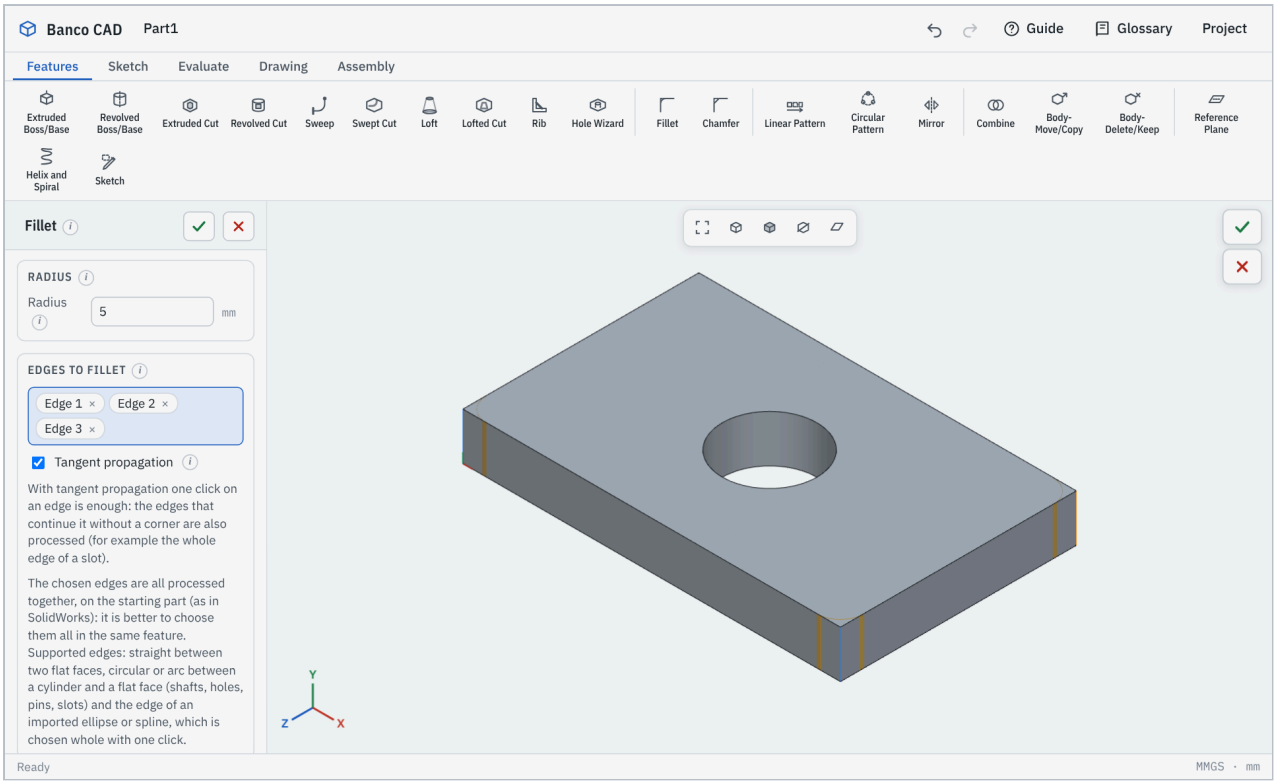
1. In the **Features** tab press **Extruded Cut**. It already starts with the **Through All** end condition, so it goes through the whole plate.
2. Confirm with **✓**.



Preview of the cut: in red the material that will be removed.

3.6 Fillet the edges

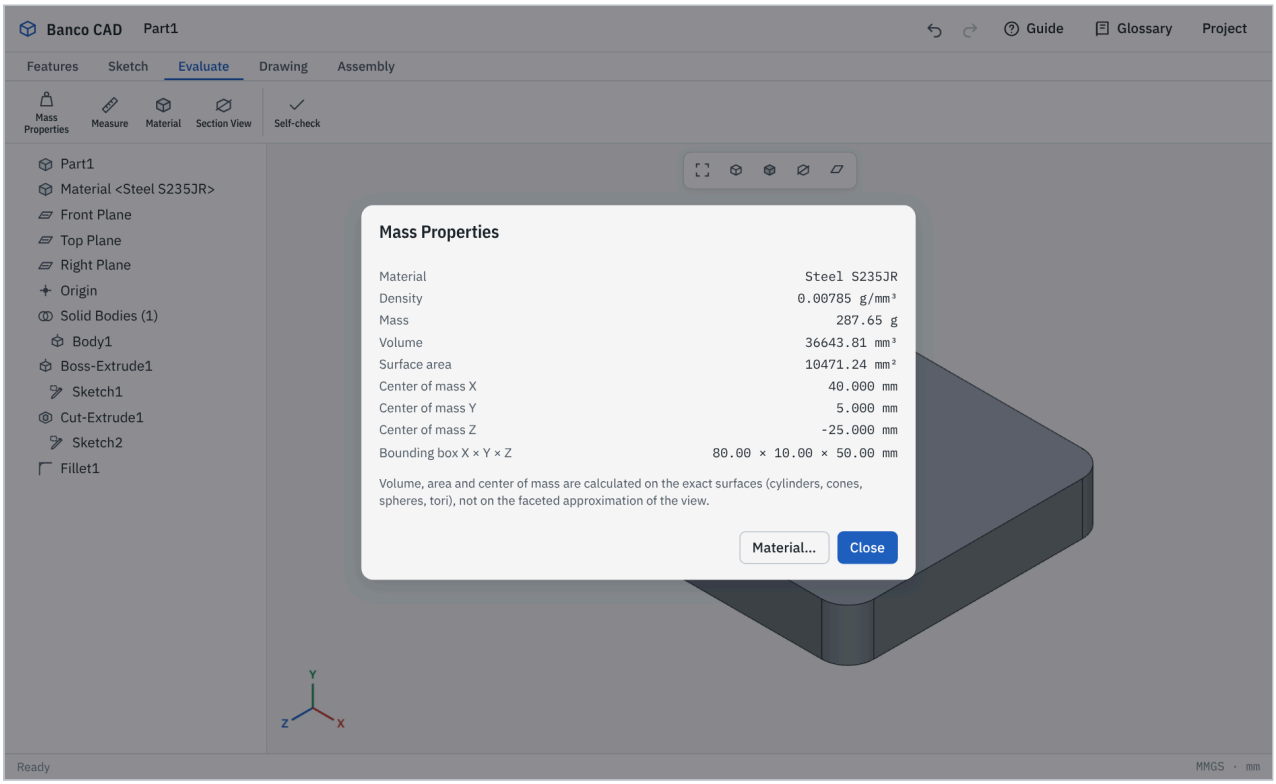
1. Press **Fillet** and type **5** in **Radius**.
2. Click three vertical edges, the ones you can see.
3. The fourth one is at the back: drag with the left button on empty space to turn the view, and click it too. The panel now lists 4 edges.
4. Confirm with ✓.



Fillet with the first three edges chosen.

3.7 Choose the material and check the numbers

1. Open the **Evaluate** tab, press **Material**, choose **Steel S235JR** and press **Apply**.
2. Press **Mass Properties**. Compare the numbers with the hand calculation below.



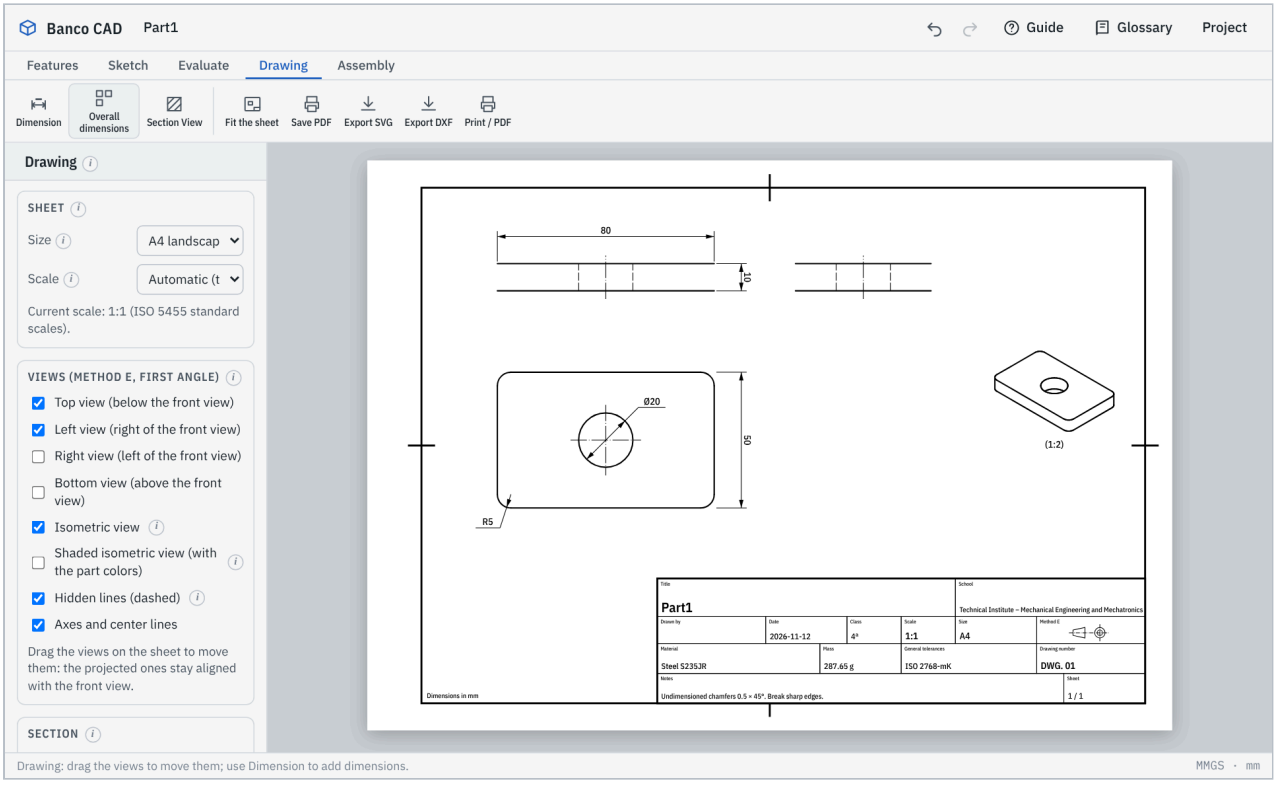
Mass properties of the finished part.

Quantity	Hand calculation	Result
Solid plate	80 × 50 × 10	40,000.00 mm³
Hole	− π × 10² × 10	− 3,141.59 mm³
4 fillets	− 4 × (1 − π/4) × 5² × 10	− 214.60 mm³
Volume		**36,643.81 mm³**
Mass	36,643.81 mm³ × 0.00785 g/mm³	**287.65 g**

The program gives the same values: volume 36643.81 mm³ and mass 287.65 g. The center of mass is in the middle of the part, at X = 40, Y = 5, Z = −25 mm.

3.8 Make the drawing

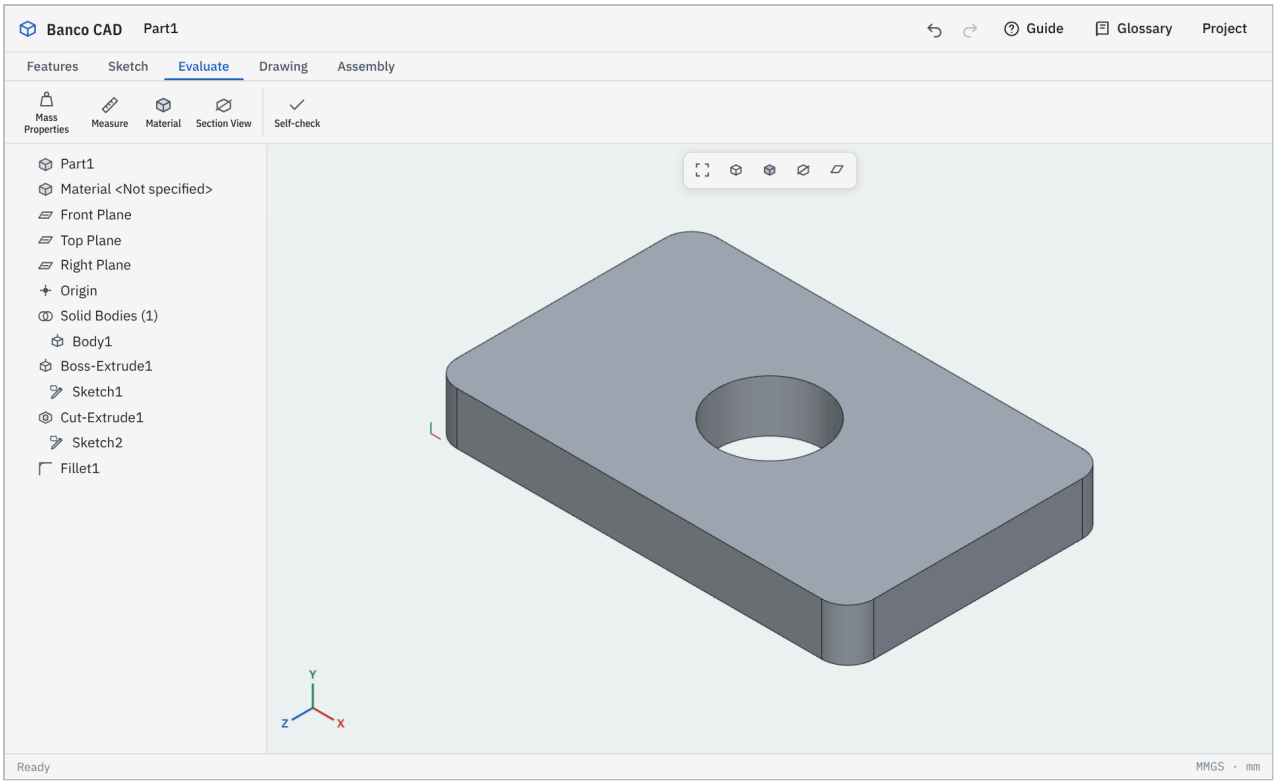
1. Open the **Drawing** tab: the views get ready by themselves, with the European method (first angle).
2. Press **Overall dimensions** to add the basic dimensions, then **Dimension** for the missing ones, like the hole.
3. Press **Save PDF** to get the drawing on the real sheet, ready to print to scale.



The drawing with views, isometric view, overall dimensions and title block.

3.9 Try changing the part

Here you see the strength of the parametric method. Double-click **Boss-Extrude1** in the tree, change the depth to 15 and confirm: hole and fillets adapt by themselves. With **Undo** you go back. From here on the manual explains every command one by one.



The finished part, with its history in the tree.

4 View, saving and importing

The basic things you need whatever you are doing: moving the view, saving your work, opening files from other programs.

4.1 Moving the view (mouse and tablet)

Look at the part from every side without changing it.

1. **Mouse – wheel:** zooms towards the point under the pointer: forward zooms in, backward zooms out.
2. **Ctrl + wheel:** rotates the model around the vertical axis.
3. **Dragging:** left button on empty space to rotate; right button (or Shift + left) to pan.
4. **Tablet:** spreading or pinching two fingers zooms. To rotate and pan, **touch the part first** (the view border turns blue and «Part held» appears): then one finger rotates and two fingers pan. Touch empty space or «Release» to let it go.
5. **Keyboard:** space bar for the orientations (Front, Top, Isometric...), F to fit the part to the window.

> **Tip** If you get lost, press Fit (the first button of the small bar at the top) and then Isometric.

4.2 Your first part in 6 steps

The basic path of every model: dimensioned sketch → feature.

Where to find it: Sketch tab › Sketch

1. **Sketch** tab › **Sketch**, then choose the Top Plane (for a plate) or the Front Plane.
2. Draw a **Corner Rectangle** with one corner on the Origin (the pointer snaps to it by itself).
3. With Smart Dimension click a side, click where to place the dimension and type the value; repeat for the other side.
4. When the sketch turns **black** it is Fully Defined.
5. Press **Extruded Boss/Base** directly (Features tab): the sketch is confirmed automatically.
6. Type the Depth and confirm with ✓.

> **Tip** Shape first, dimensions second: there is no need to draw precisely, the dimensions set the sizes.

4.3 Viewing the licenses and the authors

Know what you can do with the program and who wrote its parts.

1. Open the **Project** menu, at the top right.
2. Press **About and licenses**: you will find the GNU GPL 3 license, the parts by other authors, the trademarks mentioned and the links to the texts.

> **Tip** The program collects no data: your work stays in your browser.

4.4 Using the program in English

Work with the command names of the English version of SolidWorks, or let people who do not speak Italian use Banco CAD.

1. Choose **Progetto › Language: English**: your work is saved and the page reopens in English, with the same project.
2. To go back to Italian choose **Project › Lingua: italiano**.
3. In English numbers are written with a decimal point (12.5); in the fields you can also type a comma.

> **Tip** The choice stays in this browser. A link to the site with ?lang=en at the end of the address opens the English version directly.

> **Tip** The names of features made in Italian (Estrusione1) are shown in English (Boss-Extrude1) without changing the project; a name you typed yourself stays as it is.

4.5 Saving, exporting and reopening your work

Never lose your work and take it to another computer.

1. Your work **saves itself** in this browser at every change (even if you close the page).
2. To move it: **Project › Export project** and save the file (or copy the text).
3. On another computer: **Project › Import a project** and choose the file (or paste the text).
4. **Project › New project** clears the bench: export first if you want to keep the work done.
5. **Project › Export STEP** saves the part to open it in another CAD program (SolidWorks, Inventor, Fusion, FreeCAD...).
6. **Project › Export STL** creates the file for 3D printing.
7. **Project › Save PDF** creates a PDF of what you are looking at: the drawing, the open sketch or the 3D view.

> **Tip** Automatic saving works for this browser and this device: to hand in a piece of work, use the export.

> **Tip** The project file (Export project) can only be reopened by Banco CAD; for other programs use STEP (3D part), DXF (2D drawing or sketch) or PDF (to read and print).

4.6 Importing a part from an STL or OBJ file

Use a part made with another program (SolidWorks, Inventor, Tinkercad...) to measure it, put it on a drawing or modify it.

1. In the original program save the part as STL (or OBJ). In SolidWorks: File › Save As › type STL. If the program can write STEP, use that: see «Importing a part from a STEP or IGES file».
2. Here: **Project › Import STL or OBJ** and choose the file.
3. Choose the File units (usually millimeters) and, if the part arrives lying on its side, check File Z axis up. Press **Import**.
4. **Imported1** appears in the tree: it is a Faceted body. You can measure it, compute its mass (Evaluate tab) and put it on a drawing.

5. On its flat faces you can open a **Sketch** and add cuts, holes and extrusions, as on a part drawn here.
6. To change units or orientation: double-click **Imported1** in the tree.

> **Tip** The part arrives without its feature history: the original dimensions cannot be edited.

> **Tip** If the body not closed warning appears, the file has holes: volume and mass are not reliable.

> **Tip** At most 100,000 triangles; larger files must be simplified first.

4.7 Importing a part from a STEP or IGES file

Bring here, with its true surfaces, a part designed with SolidWorks, Inventor, Fusion, FreeCAD or another CAD program: to measure it, put it on a drawing or modify it.

1. In the original program save the part as STEP. SolidWorks: File › Save As › type «STEP AP214 (*.step)». Inventor: Export › CAD Format › STEP. If only IGES is available, that works too.
2. Here: **Project › Import STEP or IGES** and choose the file (.step, .stp, .iges or .igs).
3. The unit of measure is written in the file: you do not have to choose it. If the part arrives lying on its side, check File Z axis up. Press **Import**.
4. The first time, wait a few seconds: the reader (OpenCascade) gets ready. The file is read separately: meanwhile the page stays free, and with **Cancel** you can stop the reading.
5. If the file contains several bodies (an assembly), choose with the check boxes which ones to import (Bodies of an assembly). With **One «Imported» for each body** checked, each body becomes a feature of its own; otherwise a single **Imported1** appears in the tree.
6. The faces of the model stay recognizable: on flat faces you can open a **Sketch** and add cuts, holes and extrusions.
7. Volume, area and mass (Evaluate tab) have an error well below 1 %. To change the orientation or turn off the File colors: double-click **Imported1**.

> **Tip** The part arrives without its feature history: the original dimensions cannot be edited.

> **Tip** Each body of an assembly arrives in its own «Imported» feature (it can be suppressed or deleted on its own); with the box unchecked they all arrive in a single feature.

> **Tip** The face colors written in the file are shown on the part; the mass still uses the chosen material.

> **Tip** The .sldprt (SolidWorks) and .ipt (Inventor) files cannot be opened: save them as STEP first.

> **Tip** The reader is inside the page: it also works without the Internet.

5 The sketch

The sketch is the 2D drawing from which almost all features are made. First you draw the shape, then dimensions and relations set the sizes.

5.1 Opening a sketch (on a plane or on a face)

Every feature starts from a 2D sketch.

Where to find it: Sketch tab › Sketch

1. Select a plane in the tree (Front, Top or Right Plane) **or** a flat face of the part.
2. Press **Sketch**. If you have not selected anything, the program asks you to choose the plane.
3. The view turns Normal To the plane; the plane axes are dashed (X red, Y green, Z blue).
4. Draw, dimension, then confirm with ✓ (top right) or press a feature directly.

> **Tip** On the face of a part the edges are shown in gray: you can snap and dimension to them (External geometry).

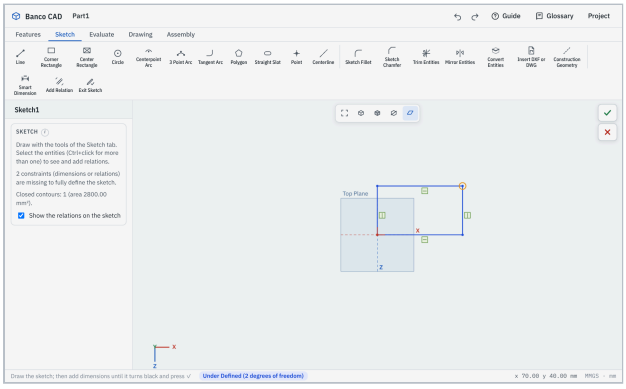
> **Tip** Right-click a face or a plane › «Sketch on this plane».

5.2 Understanding the colors: blue, black, red

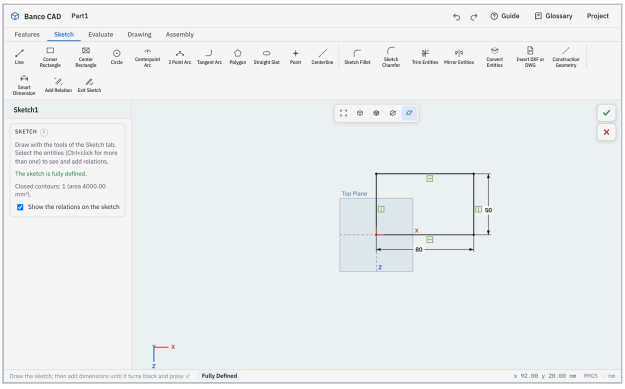
The color tells you whether the sketch is well defined.

1. **Blue:** Under Defined – some point can still be dragged; dimensions or relations are missing.
2. **Black:** Fully Defined – everything is fixed. This is the goal.
3. **Red:** Over Defined – there is one dimension or relation too many: remove the extra one: select it and press Delete, or remove it from the list of relations.
4. The status bar at the bottom repeats the state of the sketch.
5. Drag a blue entity to understand what is missing: it moves in the direction that is not yet constrained.

> **Tip** Always tie the sketch to the origin (a dimension or a Coincident relation): otherwise it stays blue.



Blue: under defined.



Black: fully defined.

5.3 Line

Outlines made of segments: L and T profiles, any polygon.

Where to find it: Sketch tab › Line

1. Press **Line**.
2. Click for the first point, then click once for each following vertex.
3. To close the outline click the first point again (it snaps).
4. Double-click or Esc to end the chain.
5. In the panel on the left you can type length and angle instead of clicking.

> **Tip** If you draw a line that is almost horizontal or almost vertical, the Horizontal or Vertical relation is added by itself.

5.4 Corner rectangle and center rectangle

Plates, blocks, pockets.

Where to find it: Sketch tab › Corner Rectangle

1. **Corner Rectangle:** click one corner, then the opposite corner.
2. **Center Rectangle:** click the center (usually the origin), then a corner: the rectangle is symmetric.
3. Dimension one horizontal side and one vertical side with Smart Dimension.

> **Tip** With the center rectangle on the origin, two dimensions are enough to make it black.

5.5 Circle

Holes, bosses, shafts, flanges.

Where to find it: Sketch tab › Circle

1. Press **Circle**.
2. Click the center, then a point on the circumference.
3. Dimension the diameter: Smart Dimension and click the circle (it becomes Ø).
4. Dimension the position of the center from the origin or from the edges.

> **Tip** Several identical circles: Equal relation, so one dimension is enough for all of them.

5.6 Arcs: centerpoint, 3 point, tangent

Fillets, slots, rounded profiles, sweep paths.

Where to find it: Sketch tab › Tangent Arc

1. **Centerpoint Arc:** click the center, the start point and the end point.
2. **3 Point Arc:** click the start, the end and a point in between.
3. **Tangent Arc:** click the end of a line (or of an arc), then the end point: the arc continues without a corner.
4. Dimension the radius with Smart Dimension (click the arc).

> **Tip** For sweep paths always use the tangent arc: the path must have no corners.

5.7 Polygon and slot

Hexagons for screw heads and nuts; slots for adjustments.

Where to find it: Sketch tab › Polygon

1. **Polygon:** choose the number of sides in the panel on the left, click the center, then a vertex.
2. **Straight Slot:** click the center of the first arc, the center of the second, then click once for the width.
3. Dimension the distance between the centers and the width.

5.8 Centerline and construction geometry

Axes of revolution and of symmetry; helper lines that do not become material.

Where to find it: Sketch tab › Centerline

1. **Centerline:** click the first and the second point. It is drawn dash-dot.
2. It is used as the axis for **Revolve**, for **Mirror Entities** and for the Symmetric relation.
3. **Construction Geometry:** select entities already drawn and press Construction Geometry: they become dashed and make no material (and the other way round).

> **Tip** For a turned part, draw the centerline on the Y axis through the origin and half of the profile on one side.

5.9 Sketch fillet and chamfer

Round or bevel a corner of the profile before extruding it.

Where to find it: Sketch tab › Sketch Fillet

1. Press **Sketch Fillet** (or **Sketch Chamfer**).
2. Type the radius (or the distance) in the panel.
3. Click the vertex shared by two lines.

> **Tip** To round the edges of a solid already made, use the Fillet on the Features tab instead.

5.10 Trim Entities

Remove the extra pieces of line where entities cross.

Where to find it: Sketch tab › Trim Entities

1. Press **Trim Entities**.
2. Click the part to remove: it disappears up to the nearest intersections.
3. Repeat until the outlines are closed and without crossings.

> **Tip** An outline that crosses itself cannot be extruded: the red message says so.

5.11 Mirror Entities

Draw half of the profile and get the other symmetric half.

Where to find it: Sketch tab › Mirror Entities

1. Draw a Centerline and half of the profile.
2. Select the entities to copy and the centerline (Ctrl+click).
3. Press **Mirror Entities**: the copy has the symmetry relations.

5.12 Importing a DXF or DWG drawing into a sketch

Use the outline of a part drawn with another program (AutoCAD, DraftSight, LibreCAD, a SolidWorks drawing...) without redrawing it.

Where to find it: Sketch tab › Insert DXF or DWG

1. Get the drawing as a DXF file or DWG file. If it has a title block, border or hatches, you can turn off their layers in the window.
2. On the tab: **Sketch › Insert DXF or DWG** (or **Project › Import DXF or DWG (sketch)**). Outside a sketch choose the **plane**; inside a sketch the geometry goes into the open one.
3. Choose the file: the window shows a **preview** of the drawing, with a small red cross on the point that will go on the origin. If the drawing has several layers, uncheck the ones you do not need (title block, dimensions, border): the preview updates.
4. Choose the File units (usually «from the file») and the **position**: as in the file, or with the lower left corner or the center on the origin. The **Texts, dimensions, hatches and tables as reference background** box brings them under the sketch, in gray (Drawing background): also the hatches, the geometric tolerances and the multileaders. Press **Import**.
5. The entities arrive selected and blue (Under Defined): check that the outlines are closed, then press ✓. Ellipses and splines arrive as chains of tangent arcs (biarcs).
6. If you have the background, read the gray sizes and put them back with **Smart Dimension**: those in the file do not drive the shape. In the sketch panel you can hide it (**Show the background**) or remove it (**Remove the background**).
7. Select the sketch and use **Extruded Boss/Base**, or **Revolved Boss/Base** if there is a Centerline.

> **Tip** Dash-dot lines in the file become centerlines, dashed lines become construction geometry (see Line type).

> **Tip** Very close end points are joined and double lines removed, so the outlines close.

> **Tip** If the part arrives 10 or 25.4 times larger or smaller, change the unit and repeat.

> **Tip** The final message says into how many arcs each ellipse or spline was divided, and the Curve deviation: usually a few thousandths of a millimeter.

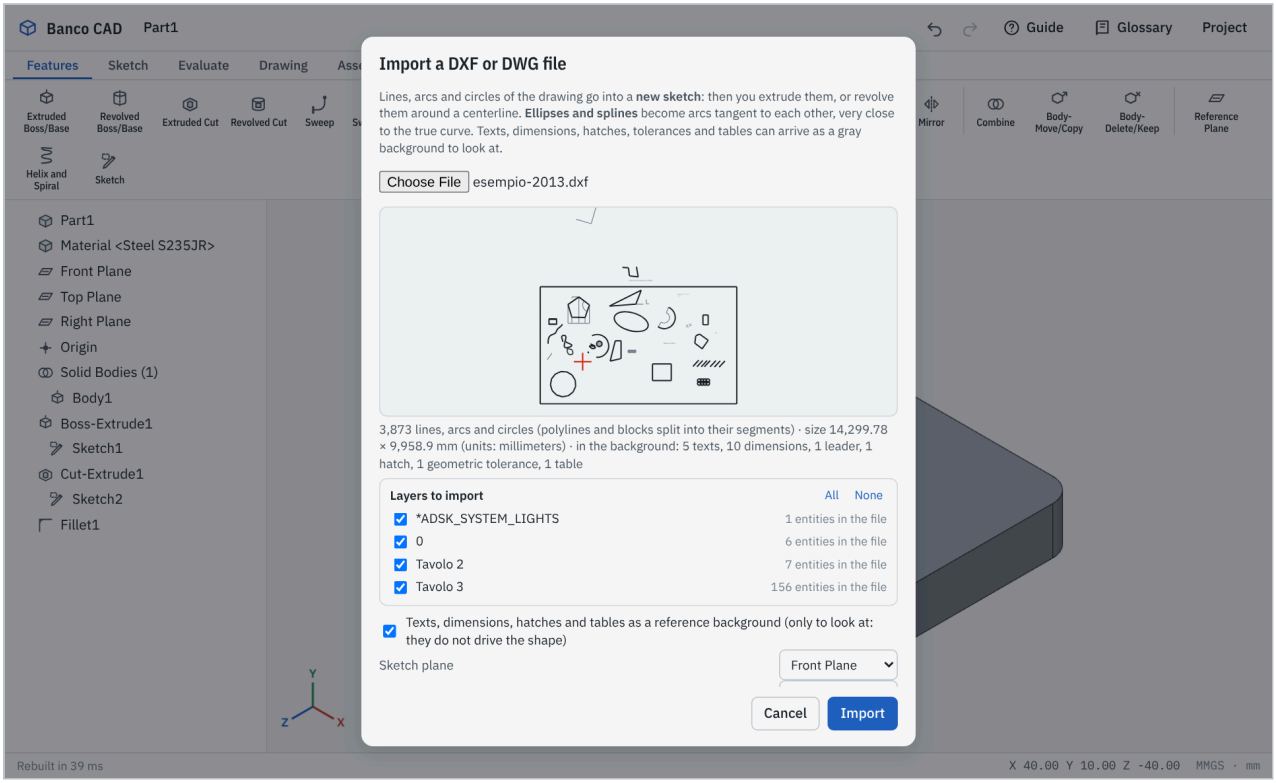
> **Tip** You can also import whole drawings with thousands of entities; to keep only the part, uncheck the layers you do not need (title block, border, dimensions).

> **Tip** Layers that are off or frozen in the file start unchecked.

> **Tip** You can also choose a **.zip** with the DXF inside: that is how the DWG → DXF Converter saves files in the published page. If the zip holds several drawings (the converter's «Save all DXF» button), choose one in **Drawing in the zip**.

> **Tip** A **DWG** takes a moment longer, because it is first turned into a DXF: meanwhile the page stays free, and Cancel stops the reading.

> **Tip** To save DXF files, for other programs or for many drawings at once, there is the DWG → DXF Converter.



The import window: preview, layers, background.

5.13 Convert Entities

Use the edges of the part in the sketch (for example for a boss on the border).

Where to find it: Sketch tab › Convert Entities

1. Open the sketch on a face: the edges of the face are gray.
2. Select the gray edges you need.
3. Press **Convert Entities**: they become sketch lines tied to the part.

5.14 Smart Dimension

Give the sizes: lengths, distances, diameters, radii, angles.

Where to find it: Sketch tab › Smart Dimension

1. Press **Smart Dimension**.
2. **Length:** click a line. **Distance:** click two points or two parallel lines. **Angle:** click two lines that are not parallel.
3. **Diameter:** click a circle. **Radius:** click an arc.

4. Click where you want the dimension, type the value (expressions like 40/2 are fine too) and press Enter.
5. To change a dimension: double-click the number.

> **Tip** If the dimension would turn the sketch red, it becomes a Driven dimension (in brackets): remove the extra constraint first.

5.15 Adding a relation

Geometric constraints: parallel, perpendicular, tangent, equal...

Where to find it: Sketch tab › Add Relation

1. Select the entities (Ctrl+click for more than one).
2. Only the possible relations appear in the panel on the left: press the one you need.
3. The existing relations are listed in the panel: the × removes them.
4. With «Show the relations on the sketch» the symbols appear next to the entities.

> **Tip** A relation in the right place is worth more than a dimension: for example Equal among four holes, or Horizontal between two centers.

6 The features

Features turn sketches into solids and work on them. Each one is added to the tree and can be edited at any time.

6.1 Extruded Boss/Base

Give thickness to a closed profile: plates, blocks, bosses.

Where to find it: Features tab › Extruded Boss/Base

1. Draw a closed sketch (holes inside are fine too).
2. Press **Extruded Boss/Base** (even while you are still in the sketch).
3. Choose the End Condition: Blind with the depth, or Mid Plane to get a part symmetric about the plane.
4. If needed: «Reverse Direction», Direction 2 to extrude to the other side too, Draft to tilt the walls.
5. Look at the orange preview and confirm with ✓.

> **Tip** The sketch to use is chosen from the «Sketch» list or by clicking its lines in the view.

6.2 Extruded Cut

Remove material: pockets, shaped holes, grooves, recesses.

Where to find it: Features tab › Extruded Cut

1. Make a sketch on the face where the cut starts (select the face › Sketch).
2. Draw and dimension the outline to remove.
3. Press **Extruded Cut**.
4. Choose Through All to go through the part, or Blind with the depth.
5. Confirm with ✓ (the preview is red).

> **Tip** For standard holes (screws, counterbores, threads) use the Hole Wizard.

6.3 Draft (tapered walls)

Molded or cast parts: tapered walls let the part come out of the mold.

Where to find it: Features tab › Extruded Boss/Base

1. Open an Extruded Boss/Base or an Extruded Cut with the Blind condition.
2. Check **Draft** and type the angle (for example 3°).
3. Normally the profile gets smaller going away from the sketch; check «Draft outward» to make it larger.
4. Confirm: flat faces become tilted planes, cylinders become cones.

> **Tip** Draft works with the Blind condition in one direction only.

6.4 Revolved boss/base and revolved cut

Turned parts: shafts, pins, bushings, knobs; grooves and slots.

Where to find it: Features tab › Revolved Boss/Base

1. Sketch on the Front Plane: a Centerline (the axis) and **half of the profile**, closed, all on one side of the axis.
2. Dimension the lengths and the distances of the points from the centerline (they are the radii of the part).
3. Press **Revolved Boss/Base** (or **Revolved Cut** to remove material).
4. Angle 360° for a full turn; less for part of a turn.
5. Confirm with ✓.

> **Tip** If the profile crosses the axis an error appears: move it all to one side.

6.5 Sweep

Bent tubes, handles, wire springs, seals: a profile that slides along a line.

Where to find it: Features tab › Sweep

1. **Path:** in a sketch draw **tangent** lines and arcs (use Tangent Arc or Sketch Fillet at the corners). Confirm.
2. **Profile:** on a plane perpendicular to the path, at the point where the path starts, draw the closed outline (for example a circle). Confirm.
3. Press **Sweep**: profile and path are chosen automatically; if needed change them from the lists or with «Pick in the view».
4. Confirm with ✓. To remove material use **Swept Cut**.

> **Tip** For springs and threads the path is a Helix and Spiral curve instead of a sketch.

> **Tip** Example: path on the Top Plane starting from the origin, profile (circle) on the Front Plane centered on the origin.

> **Tip** The radius of the path arcs must be larger than the profile.

6.6 Helix and spiral: springs and threads

Helical springs, spiral springs, real threads: a profile that slides along a curve turning around an axis.

Where to find it: Features tab › Helix and Spiral

1. **Circle:** open a sketch on the plane perpendicular to the spring axis (for example the Top Plane) and draw a Circle with the mean diameter of the spring. Confirm.
2. Press **Helix and Spiral**: in Defined by choose which sizes to give (for example Pitch 10 and Revolutions 5) and look at the orange curve. Confirm with ✓.
3. **Profile:** open a sketch on a plane through the start of the helix (with the circle on the Top Plane and Start angle 0° it is the Front Plane) and draw the wire: a circle centered on the start point of the curve. Confirm.

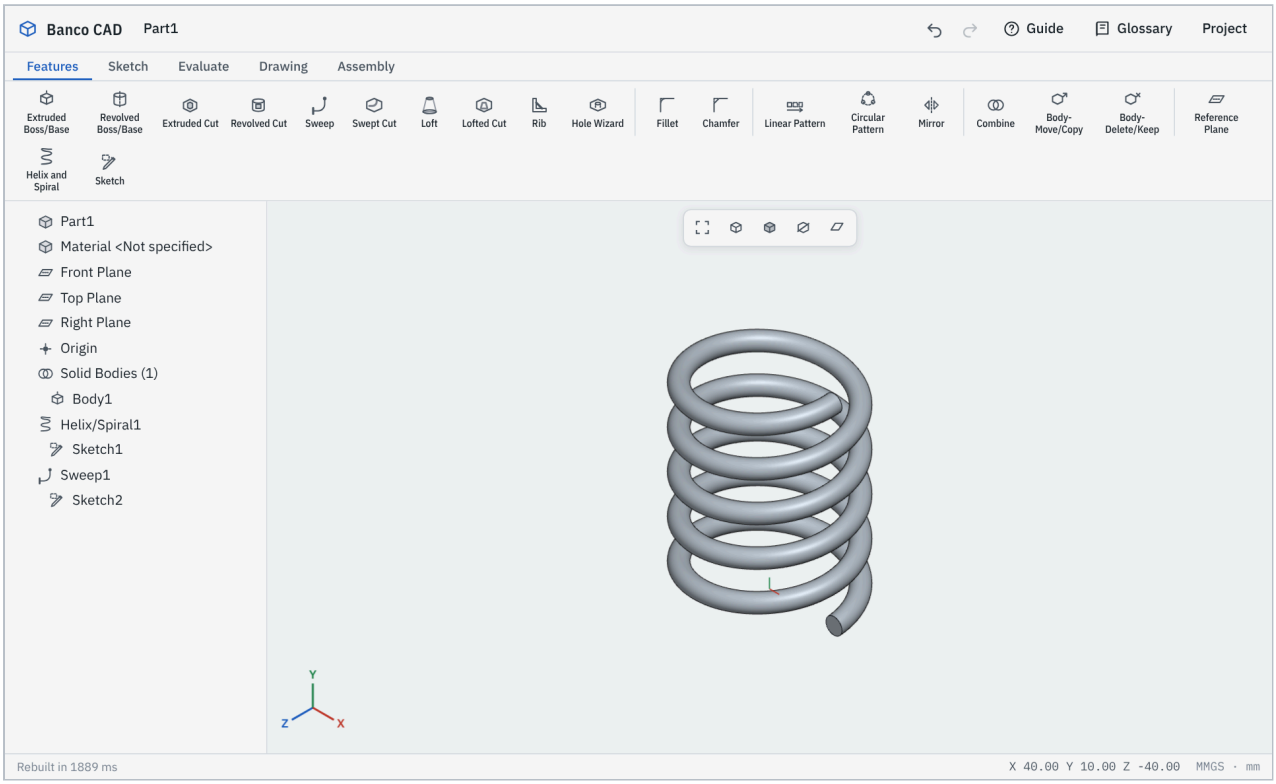
4. Press **Sweep**: the profile and the curve are chosen automatically (the curve can also be chosen from the «Path» list or by clicking it in the view). Confirm with ✓.

> **Tip** The pitch must be larger than the profile, otherwise the coils run into each other and an error appears.

> **Tip** For a conical spring check Taper Helix; for a flat spring choose «Spiral».

> **Tip** For a real thread use **Swept Cut** with a triangular profile on a cylinder.

> **Tip** The length of the wire is shown in the curve panel.



A spring: a Ø5 circle swept along a helix with 12 mm pitch and 5 revolutions.

6.7 Loft: blades and shape transitions

Solids whose section changes: twisted blades, hoppers, transitions from a square to a circle, handles.

Where to find it: Features tab › Loft

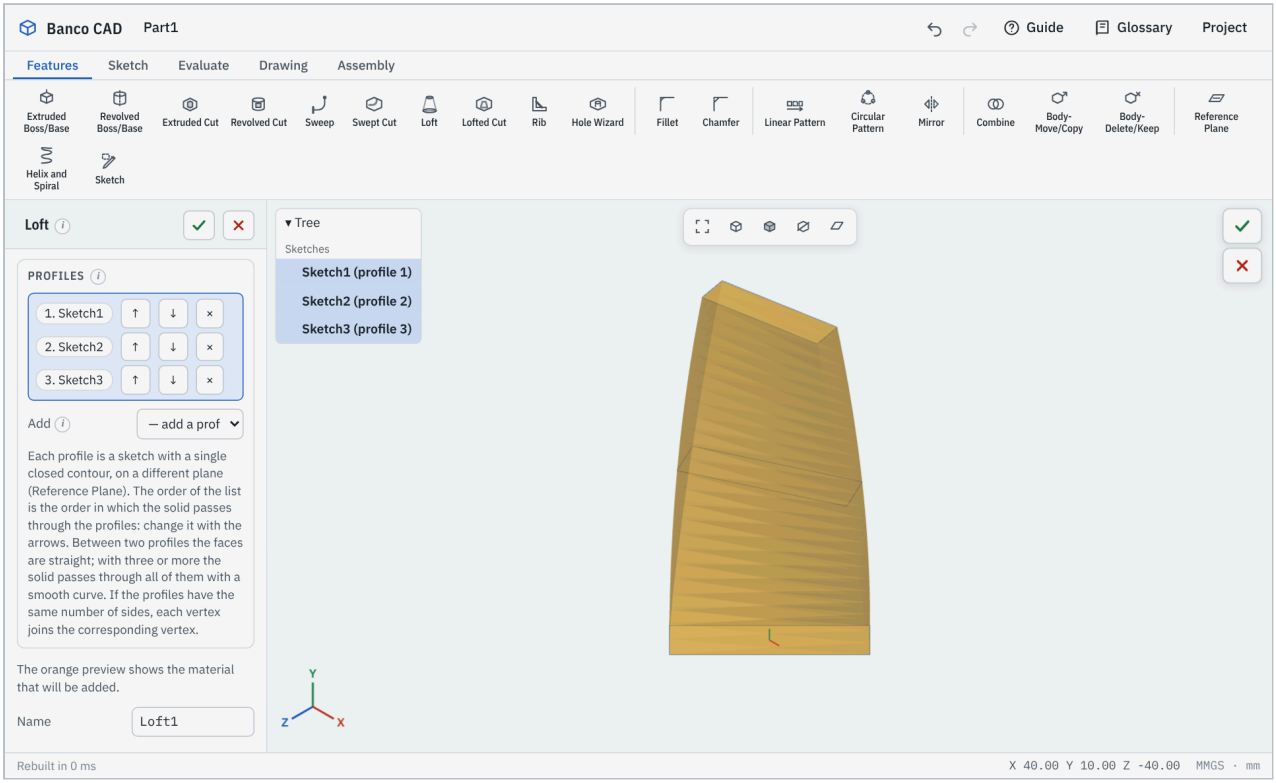
1. **First profile:** in a sketch draw the first closed outline (for example on the Top Plane). Confirm.
2. **Other planes:** with Reference Plane create a parallel plane at the distance you want, one for each other profile.
3. **Other profiles:** on each plane draw its closed outline (one outline per sketch). Confirm each time.
4. Press **Loft**: the free sketches go into the Profiles list in tree order. Click in the view to add or remove one; use the ↑ ↓ arrows to change the order.
5. Check the orange preview and confirm with ✓. To remove material use **Lofted Cut**.

> **Tip** **Twisted blade:** draw the same profile on two or three planes, each time rotated by a few degrees: the loft joins them by twisting the blade (Twist).

> **Tip** If the profiles have the same number of sides (for example two rectangles), each vertex joins its matching vertex and the faces stay separate.

> **Tip** Between a square and a circle the points are matched along the perimeter: the surface passes smoothly from one shape to the other.

> **Tip** Two profiles on the same plane do not work: each profile needs its own plane.



Lofted Boss/Base: three rotated profiles on three different planes become a twisted blade.

6.8 Rib

Thin stiffening walls, for example in the corner of a bracket.

Where to find it: Features tab › Rib

1. Open a sketch on a plane that cuts the part where the rib goes (often the Front Plane or a mid plane).
2. Draw **just one line** (open) where the free edge of the rib must run. It does not need to touch the part: it is extended automatically.
3. Press **Rib**, type the Thickness and choose the side (mid plane or one side).
4. If the material goes to the wrong side use «Flip material side». Confirm with ✓.

6.9 Hole Wizard

Holes for screws according to the standards: clearance, counterbored, countersunk, tapped.

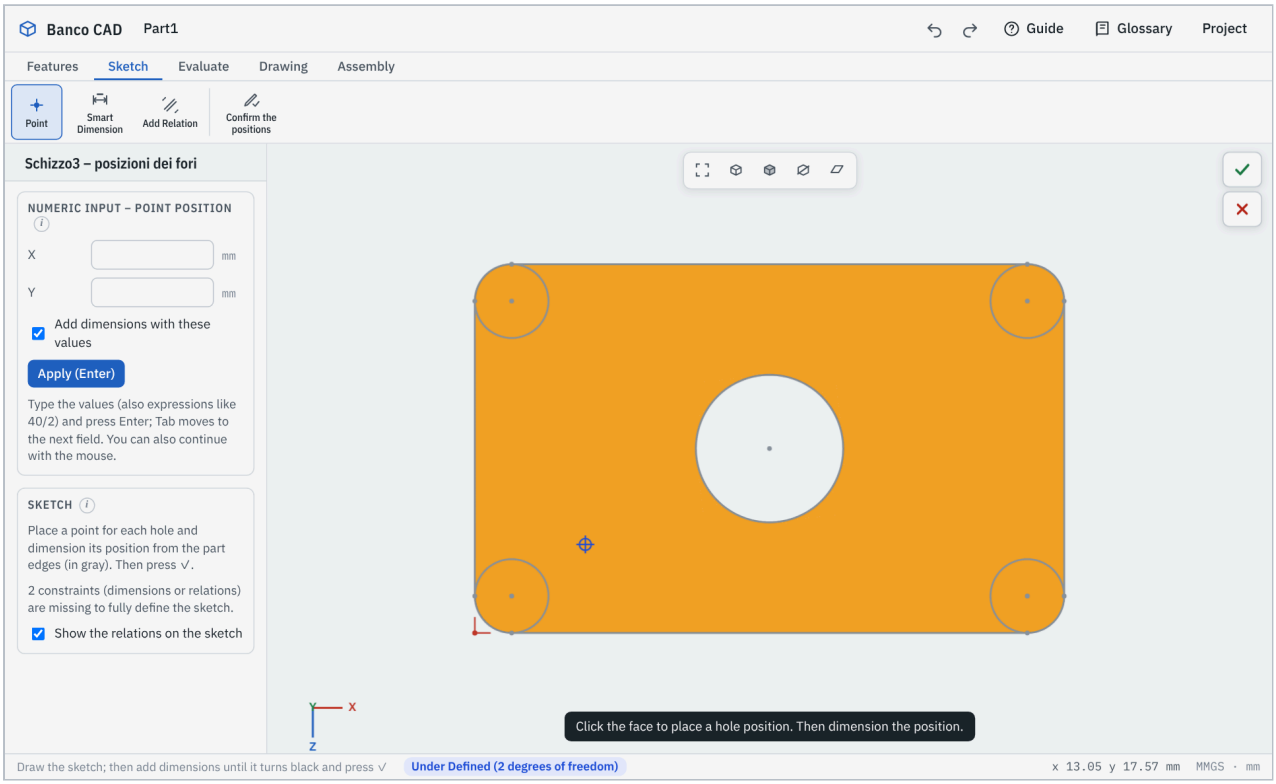
Where to find it: Features tab › Hole Wizard

1. Press **Hole Wizard** and click the flat face where the holes go.

2. The sketch of the positions opens: click once for each center, then dimension the positions (for example from the gray edges). Confirm with ✓.
3. In the panel choose the Hole type, the Size (M3…M20) and the fit.
4. Choose Through All, or Blind with the depth. The standard diameters appear under Standard values.
5. Confirm with ✓.

> **Tip** The depth of blind holes is measured up to the start of the 118° drill point.

> **Tip** To change the positions: double-click the hole › «Edit the positions».



Hole Wizard: the face in orange, then you place the positions.

6.10 Working with several solid bodies

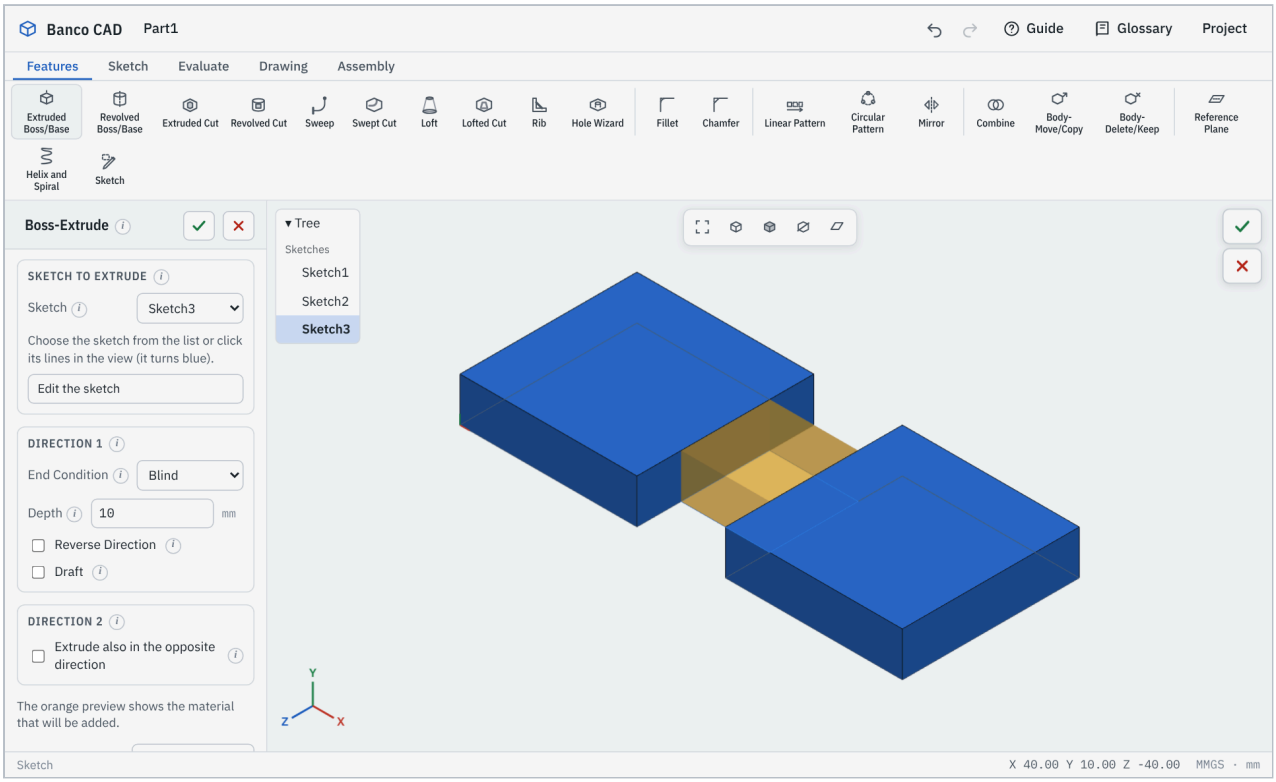
Keep separate solids inside the same part, to combine them later.

Where to find it: Features tab › Extruded Boss/Base

1. Create the first solid, then a new feature that adds material (**Extruded Boss/Base**, **Revolved Boss/Base**, **Sweep** or an imported body).
2. In the panel uncheck **Merge result**: a new body is born, and it appears in the **Solid Bodies** folder of the tree.
3. With several bodies a new feature starts from **Feature Scope › Selected bodies** with **Auto-select** checked: it works on the bodies it touches. Uncheck it to choose them by hand, or choose **All bodies**. **Fillets and chamfers** work only on the body of the edge.
4. To repeat a whole body, in **Linear Pattern**, **Circular Pattern** or **Mirror** use the **Bodies to pattern** box (in the mirror **Bodies to mirror**, with **Merge solids** if you want a single body). While the panel is open, bodies and sketches can also be picked by name in the **tree** above the view.

5. In **Evaluate › Mass Properties** you find the data of each body; **Export STEP** writes one solid for each body.

> **Tip** With «Merge result» checked the material joins the bodies it touches; if there is more than one body and it touches none, a new body is born.



Feature Scope and the tree over the view, with sketches and bodies chosen by name.

6.11 Combining bodies: Add, Subtract, Common

Make a part by joining, subtracting or intersecting bodies, like «Combine» in SolidWorks.

Where to find it: Features tab › Combine

1. Press **Combine** and choose the operation: **Add**, **Subtract** or **Common**.
2. Check the bodies in the panel, or click them in the view (you can also pick them in the tree with Ctrl + click before pressing **Combine**). For **Subtract** first choose the **main body**, then the ones to take away from it.
3. Confirm with ✓: a single body remains.

> **Tip** If the bodies have no material in common, Common gives an error: check their position.

6.12 Moving, rotating or copying a body

Put a body in its place or make copies of it, without redoing the sketches.

Where to find it: Features tab › Body-Move/Copy

1. Press **Body-Move/Copy** and check the bodies.
2. Type the **rotation** (X, Y or Z axis through the origin and angle) and the **translation** in X, Y and Z. It rotates first, then moves.

3. For copies check **Copy the body** and type the **number of copies**: each one is moved from the previous one.

6.13 Deleting or keeping bodies

Remove the bodies you no longer need, for example those used for construction.

Where to find it: Features tab › Body-Delete/Keep

1. Press **Body-Delete/Keep** and choose whether the chosen bodies are **to delete** or **to keep**.
2. Check the bodies and confirm with ✓.

> **Tip** It is a feature in the tree: if you suppress or delete it, the bodies come back.

6.14 Fillet and chamfer of edges

Round or bevel the edges of the solid.

Where to find it: Features tab › Fillet

1. Press **Fillet** (or **Chamfer**).
2. Click **all** the edges to work: they turn blue (a second click removes them). With **Tangent propagation** checked (it already is), one click is enough for a chain of edges that continue without a corner, for example the edge of a slot.
3. Type the radius (for the chamfer: distance, or distance and angle, or two distances).
4. Confirm with ✓: the edges are worked **together**. Where two fillets meet they blend into each other; where three meet at right angles the corner becomes a piece of a sphere.

> **Tip** Put edges that touch each other in the same feature: the result is the one of SolidWorks.

> **Tip** On an inside (concave) edge the fillet adds material.

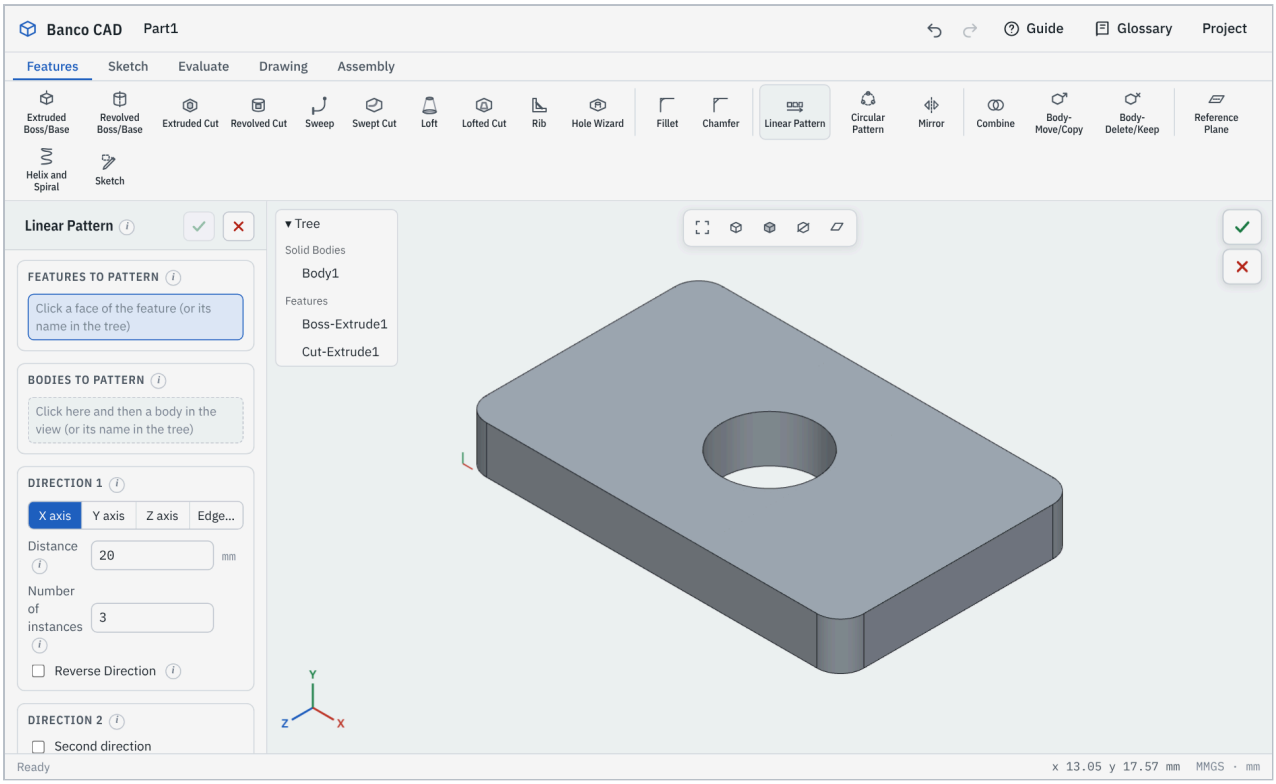
> **Tip** The edge of an elliptical or spline hole is a **curved edge**: one click is enough to pick all of it. The radius must be smaller than the radius of the curve at its tightest point.

6.15 Linear Pattern

Repeat holes or bosses in a row or in a grid.

Where to find it: Features tab › Linear Pattern

1. Press **Linear Pattern**.
2. In «Features to pattern» click a face of the feature to copy (or its name in the tree).
3. Direction 1: X, Y, Z axis or an edge of the part; type the distance (spacing) and the number of Instances.
4. For a grid check Direction 2 and fill it in the same way. Confirm with ✓.



The linear pattern panel.

6.16 Circular Pattern

Flange holes, spokes, teeth spread around an axis.

Where to find it: Features tab › Circular Pattern

1. Make the first hole (or the first boss).
2. Press **Circular Pattern** and choose the feature to repeat.
3. Axis: X, Y, Z or «From the part» and click a cylindrical face (its axis is used).
4. Number of instances and total angle (360° = spread over the whole turn). Confirm.

6.17 Mirror

Copy features symmetrically about a plane.

Where to find it: Features tab › Mirror

1. Press **Mirror**.
2. Choose the features to mirror (click one of their faces).
3. Choose the plane of symmetry: front, top, right or a flat face.
4. Confirm with ✓.

> **Tip** If the part is centered on the origin, the standard planes are already planes of symmetry.

6.18 Reference Plane

Make sketches where there is no face: in the middle of the part, higher up, offset.

Where to find it: Features tab › Reference Plane

1. Select a plane or a flat face, then press **Reference Plane**.
2. Type the distance; «Reverse» for the other side.
3. Confirm: the new plane appears in the tree and can be used for sketches.
4. **Sketch on the plane:** click its name in the tree (it turns blue in the view) and press **Sketch**. Or press **Sketch** without choosing anything and pick the plane in the panel.

7 Editing the model

The tree is the history of the part. From here you go back to any step to change it.

7.1 Editing a feature or a sketch

The model is parametric: change a size and the part updates.

1. **Double-click** the feature in the tree (or a face of the part): its panel opens again.
2. For the sketch: right-click the feature › «Edit Sketch» (or double-click the sketch under the feature).
3. Change the values and confirm: all the features after it are recalculated (Rebuild).
4. If a feature turns red it has an error: hover over it with the pointer to read why.

7.2 Suppress, rename, delete, rollback bar

Organize the history of the part.

1. Right-clicking a feature in the tree opens the menu.
2. **Suppress:** the feature stays but is not calculated (struck through). Useful for trying variants.
3. **Rename:** give clear names (e.g. «M6 fixing hole»).
4. **Roll back here:** the orange Rollback bar hides the features below it; you can insert a feature in the middle. «Roll to End» to go back.
5. **Delete:** removes the feature (and those that depend on it, after a confirmation).
6. Ctrl+Z undoes, Ctrl+Y redoes.

8 Evaluating the part

The checks on the part: mass, measurements, sections, and the proof that the program's calculations are right.

8.1 Mass properties and material

Know the volume, area, mass and center of mass of the part.

Where to find it: Evaluate tab › Mass Properties

1. **Evaluate › Material** tab: choose the material (steel, aluminum, brass, plastics...).
2. **Evaluate › Mass Properties**: volume, area, mass, center of mass and bounding box appear.
3. With «Material...» in the same window you change the material on the fly.

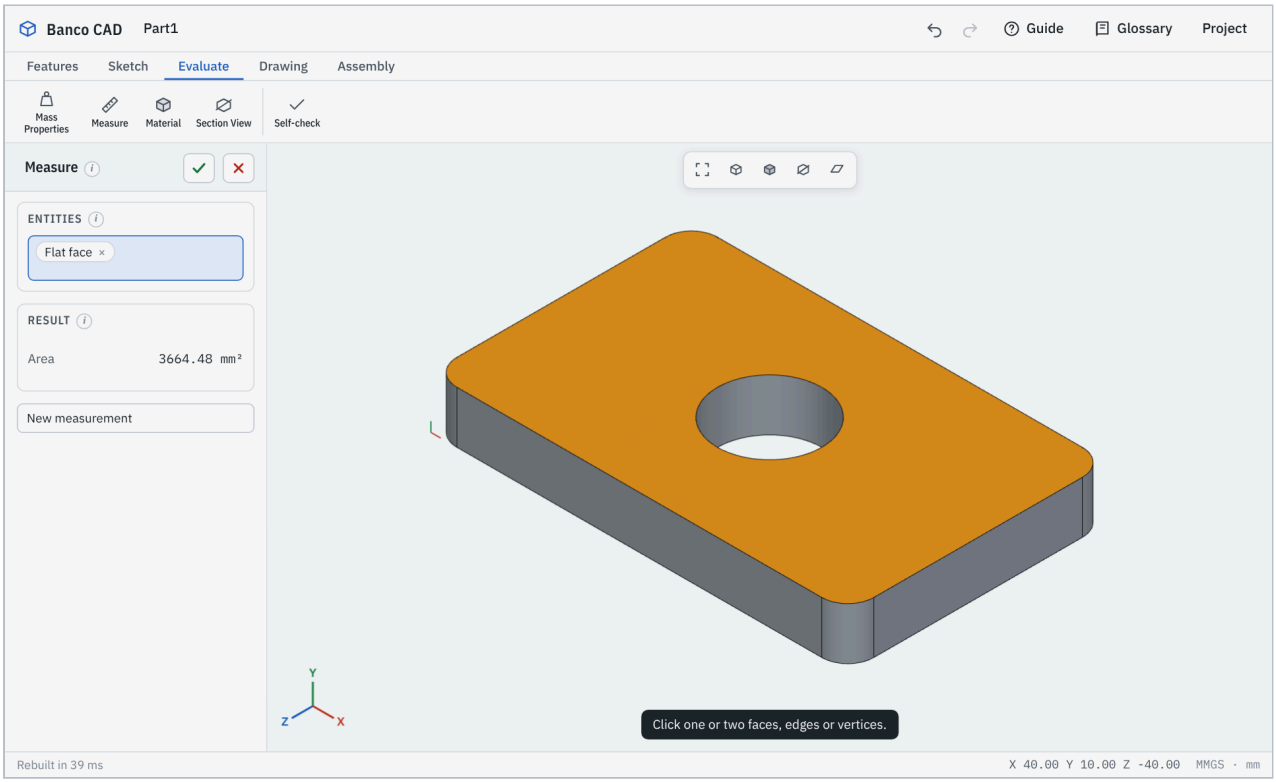
> **Tip** Without a material the mass is calculated with a density of 1 g/cm³.

8.2 Measuring

Check distances, diameters and angles on the part.

Where to find it: Evaluate tab › Measure

1. Press **Measure**.
2. Click one or two faces, edges or vertices.
3. The result appears in the panel; «New measurement» starts again.



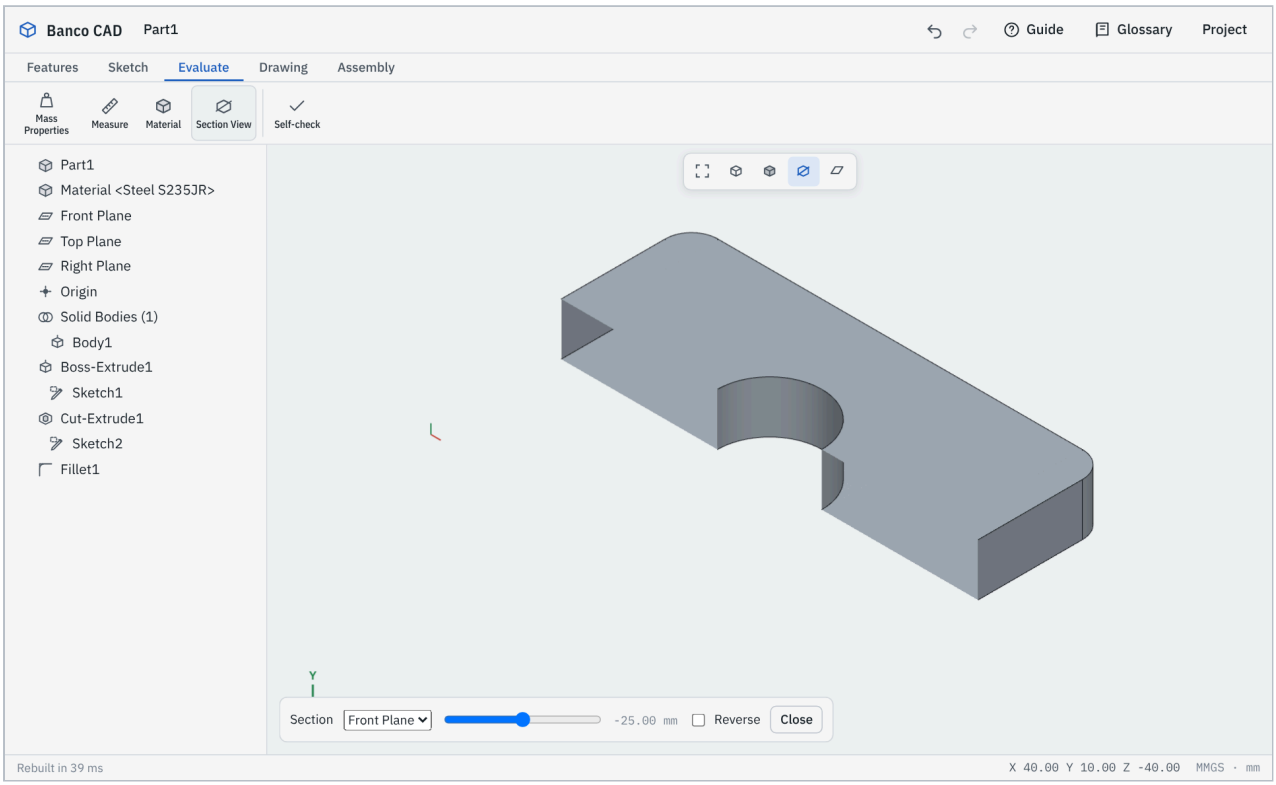
Measuring a face: the area is in the panel.

8.3 Section view (in the 3D model)

See inside the part without changing it.

Where to find it: Evaluate tab › Section View

1. Press **Section View** (also from the small bar of the view).
2. At the bottom choose the plane and move the slider to move the cut; «Reverse» changes the part that is removed.
3. «Close» removes the section.



Section view in the 3D model, with the bar at the bottom.

8.4 Program self-check

Check that the program's calculations are right.

Where to find it: Evaluate tab › Self-check

1. Press **Self-check**.
2. The program rebuilds the test parts and compares volume, area, center of mass and bounding box with the values calculated by hand.
3. The table shows, for each test, the largest error (allowed up to 1 %).

9 The assembly

The assembly puts together several parts of the same project and sets how they are connected. The parts all live inside the project, in a single file.

9.1 Creating an assembly and inserting the parts

Put several parts together, each in its place. Everything stays in a single project, that is in a single file.

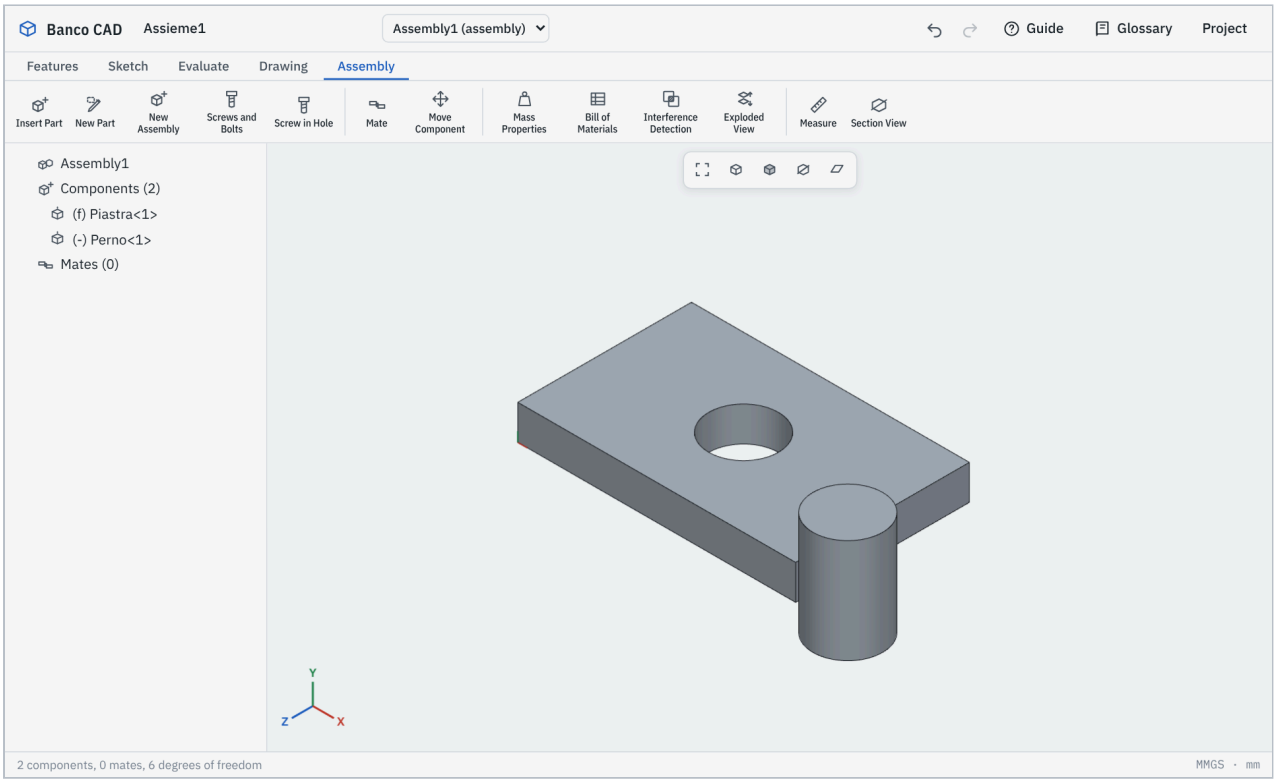
Where to find it: Assembly tab › Insert Part

1. Open the **Assembly** tab.
2. Press **Insert Part** and choose a part of the project, or a file (STEP, IGES, STL, OBJ or an exported project). The file is copied into the project.
3. The first part inserted is fixed: in the tree it has (f) before its name. The others are free and show (-) as long as they still have degrees of freedom.
4. To add another copy of the same part, insert it again or use **Copy component** from the right-click menu.

> **Tip** A STEP file with several bodies becomes one part for each body, in the positions of the file.

> **Tip** Fix or Float a component with the right button. If you drag a fixed component, only the view turns.

> **Tip** The copies are the same part: if you change the part, they all change together.



Assembly with two components inserted: (f) fixed, (-) under defined.

9.2 Several parts in the same project

Draw the parts of an assembly without leaving the project, and change them when needed.

Where to find it: Assembly tab › New Part

1. Press **New Part** on the Assembly tab, or **Project › Add a part**. The new part opens empty: draw it as usual.
2. To switch from one part to another use the list next to the name, at the top.
3. In the assembly, double-clicking a component opens its part. When you go back to the **Assembly** tab, the components update and the mates are recalculated.

> **Tip** Undo works separately for each part and for the assembly.

> **Tip** **Project › Delete this part** also removes its components and cannot be undone.

> **Tip** **Project › New project** starts again from scratch: export first if you need the work done.

> **Tip** **Project › Export project** saves all the parts and the assembly in a single file.

9.3 Several assemblies and subassemblies

Divide a machine into groups: you assemble the group once and reuse it whole.

Where to find it: Assembly tab › New Assembly

1. Press **New Assembly** and assemble the group, as usual.
2. Go back to the other assembly with the list next to the name.
3. Press **Insert Part**: the list also contains the group, marked «subassembly». Mate it like a part: choose the faces of its pieces.

> **Tip** A subassembly is rigid, until you choose Make flexible with the right button.

> **Tip** The bill of materials opens the subassemblies and counts the real parts.

> **Tip** Interference Detection also checks the pieces inside the subassembly; with the «Treat subassemblies as components» box it checks it as one block.

> **Tip** An assembly cannot contain itself: it does not appear in the list.

> **Tip** Right-click the name of the assembly in the tree to delete it.

9.4 Letting the pieces of a subassembly move

Try a mechanism mounted inside the machine: a slide moves, a hinge turns, and the mates of the upper assembly can stop it.

Where to find it: Assembly tab › Mate

1. Insert the subassembly with **Insert Part**.
2. Right-click the component in the tree: **Make flexible**.
3. Drag one of its pieces with the mouse, or mate a face of one of its pieces to another component.

> **Tip** Each copy has its own positions: the original assembly stays as it is.

> **Tip** **Make rigid** puts the pieces back where they are in their own assembly.

> **Tip** A subassembly inside a flexible one stays rigid.

9.5 Making two surfaces touch

Rest a cylinder, a sphere or a free-form shape on a plane, or a cam on a follower.

Where to find it: Assembly tab › Mate

1. Press **Mate** and choose the two elements: a plane, a cylinder or a sphere; with a free-form surface of a STEP file, a plane or a sphere.
2. Press **Tangent**. With a free-form surface it is already chosen.
3. With **Flip side** choose on which side they touch. Press ✓.

> **Tip** On free-form surfaces the precision is that of the triangles of the file.

> **Tip** Cones and edges have no Tangent mate.

9.6 Putting a screw in with one click on the hole

Mount the fasteners without making the mates by hand.

Where to find it: Assembly tab › Screw in Hole

1. Press **Screw in Hole**.
2. Click the wall of the hole, near the opening the screw goes in from.
3. In the window check type, size and length, then press **Insert**.

> **Tip** For a nut or a washer, click near the opening on the side where they go.

> **Tip** In a counterbored hole click the small hole, so the head rests in the counterbore.

> **Tip** A single Undo removes the piece and the two mates.

9.7 Putting a screw in a hole

Mount the parts with standard fasteners, without drawing them, and have them in the bill of materials.

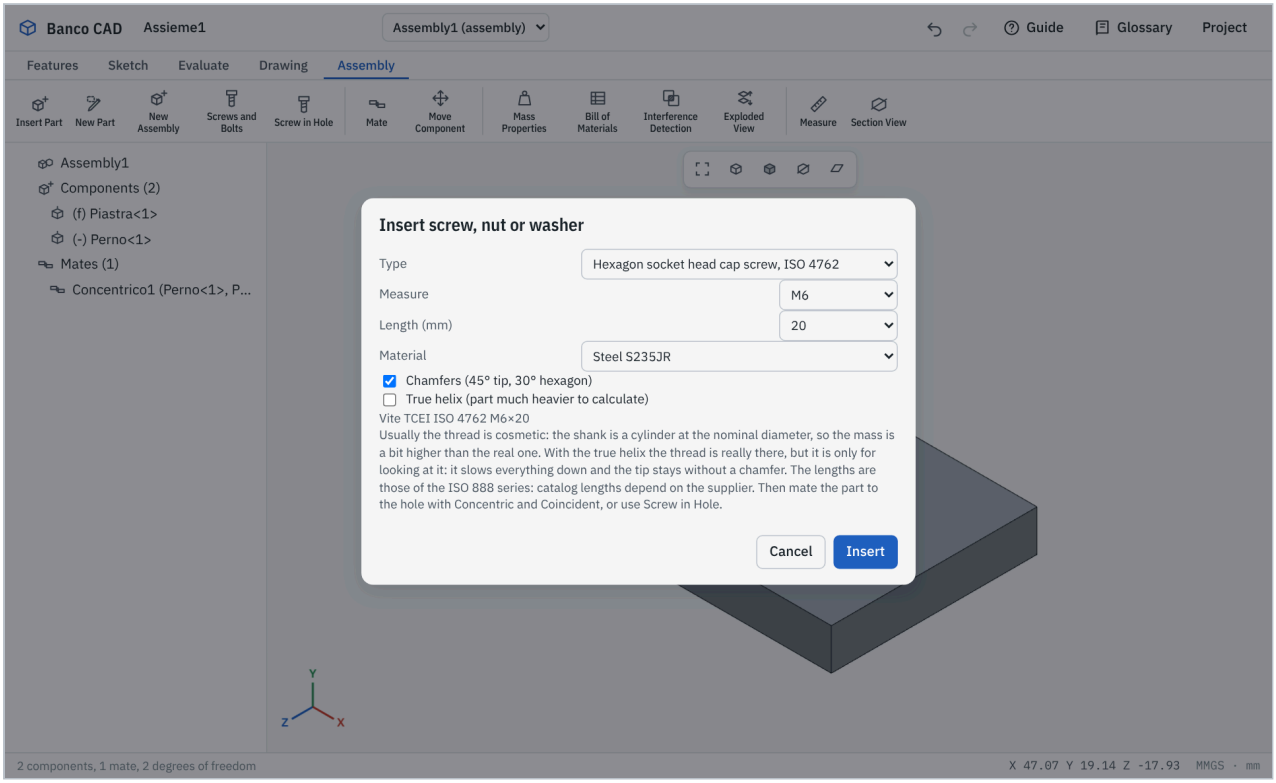
Where to find it: Assembly tab › Screws and Bolts

1. Prepare the hole in the part with the Hole Wizard: clearance, counterbored, countersunk or tapped, of the same size as the screw.
2. In the assembly press **Screws and Bolts**, choose type, size, length and material, then **Insert**.
3. Press **Mate**: **Concentric** between the shank and the hole, then **Coincident** between the underside of the head and the face of the part.
4. For each identical screw, insert it again: the bill of materials keeps a single row, with the right quantity.

> **Tip** A screw in a tapped hole or in a nut shows up in Interference Detection marked «thread»: this is normal, because the thread is only indicated.

> **Tip** On the drawing, threads are drawn with the thin lines of ISO 6410, in section views too: there the hatching goes up to the thick line.

> **Tip** A countersunk screw rests on its cone: use Coincident between the top face of the head and the face of the part, with a countersunk hole of the same size.



The window to insert screws, nuts and washers.

9.8 Mating two components

Say how two components sit relative to each other: for example a pin in a hole with its head resting on the plate.

Where to find it: Assembly tab › Mate

1. Press **Mate**.
2. Click two elements of two different components: faces, edges or vertices. The free component moves at once.
3. Check the type: Coincident, Concentric, Parallel, Perpendicular, Distance or Angle. Only those possible with the chosen elements are active.
4. If the part is turned the wrong way press **Flip alignment** (or **Flip side**, **Flip angle**).
5. Confirm with ✓. The mate appears in the tree: double-click to edit it, right-click to suppress or delete it.

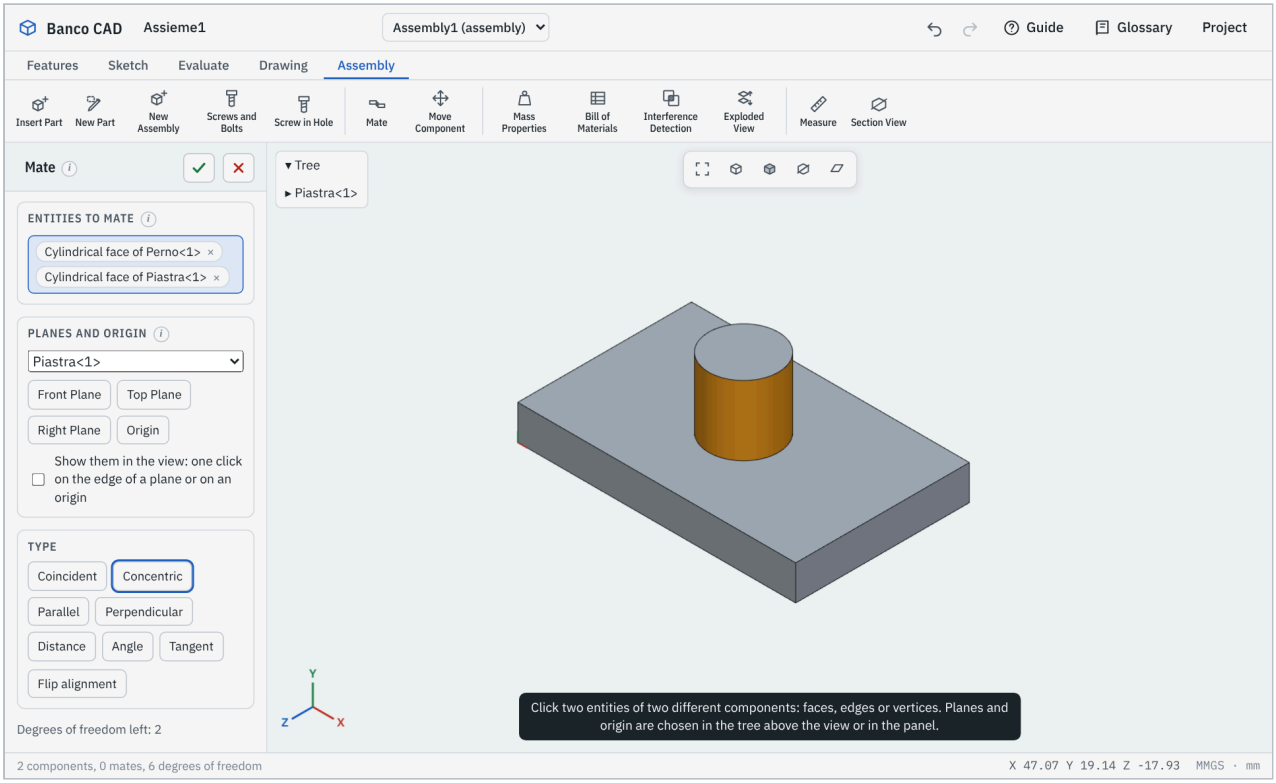
> **Tip** Coincident between two circular edges brings the two circles into contact, with the same center: handy for a screw in a hole.

Tip If two mates contradict each other, the message is red and ✓ does not confirm. A mate that becomes impossible after a change stays in the tree with an exclamation mark.

Tip Planes and origin: in the panel choose the component and press the plane or **Origin**. No faces are needed: a part can be placed on the plane of another.

Tip When you change the sizes of a part, a chosen vertex is found again by itself. The exclamation mark stays only if it really disappears, for example after a chamfer right on that vertex.

Tip Among the curved faces of a STEP file, cylinders, cones, spheres and tori are recognized. A free-form surface can only be used with **Tangent**, against a plane or a sphere: for the other mates choose an edge or a vertex.



Concentric mate between pin and hole.

9.9 Moving a component

Try out the movement of the mechanism or bring the parts closer before mating them.

Where to find it: Assembly tab › Move Component

1. With the mouse: drag the component with the left button. It moves only in the degrees of freedom it has left, and the mates stay true.
2. To rotate, or on a tablet: press **Move Component**, choose **Move** or **Rotate** and drag the component.
3. To turn the view in the assembly, drag on empty space or on a fixed component, or hold down the wheel.

Tip Undo puts the component back where it was.

Tip A component tied to others takes them along, if they can follow it.

9.10 Mass and bill of materials of an assembly

Know how much the assembled machine weighs, where its center of mass is and which parts are needed to build it.

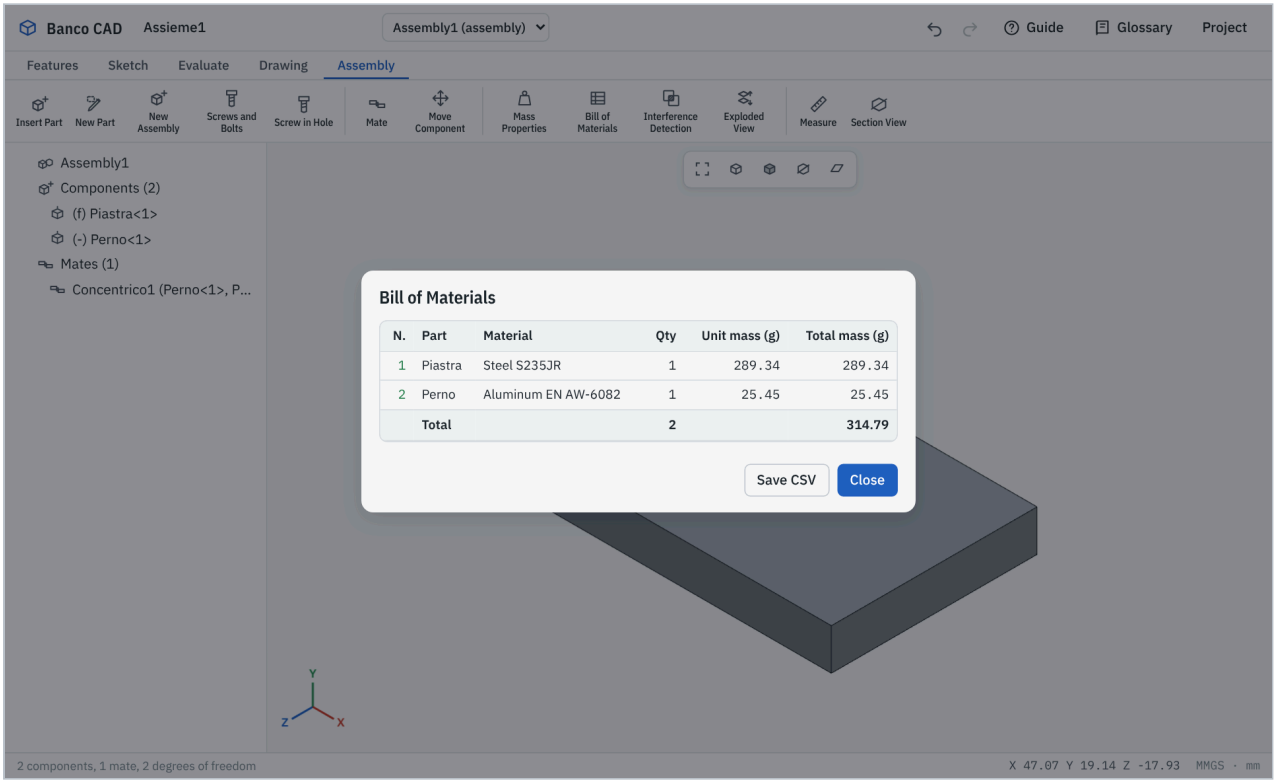
Where to find it: Assembly tab › Mass Properties

1. Give each part its material: open the part and use **Material** on the Evaluate tab.
2. On the Assembly tab press **Mass Properties**: mass, volume, area, center of mass and bounding box of the whole assembly, then one row for each component.
3. Press **Bill of Materials**: one row for each part, with quantity, material, mass of one piece and total mass.

> **Tip** The center of mass is weighted by mass: a steel part counts almost three times as much as an identical aluminum one.

> **Tip** A part without material is calculated with a density of 1 g/cm³, and the window says so.

> **Tip** Components of the same part make a single row of the bill of materials.



The bill of materials, which can be saved as CSV.

9.11 Making the exploded view

Show how the machine is put together, with the pieces moved apart along a direction.

Where to find it: Assembly tab › Exploded View

1. Press **Exploded View**.
2. Click the components to move, choose the direction **X**, **Y** or **Z** (**Reverse** for the opposite way) and type the distance, or drag a chosen component.
3. Press **Add the step** and go on with the other components, then ✓.

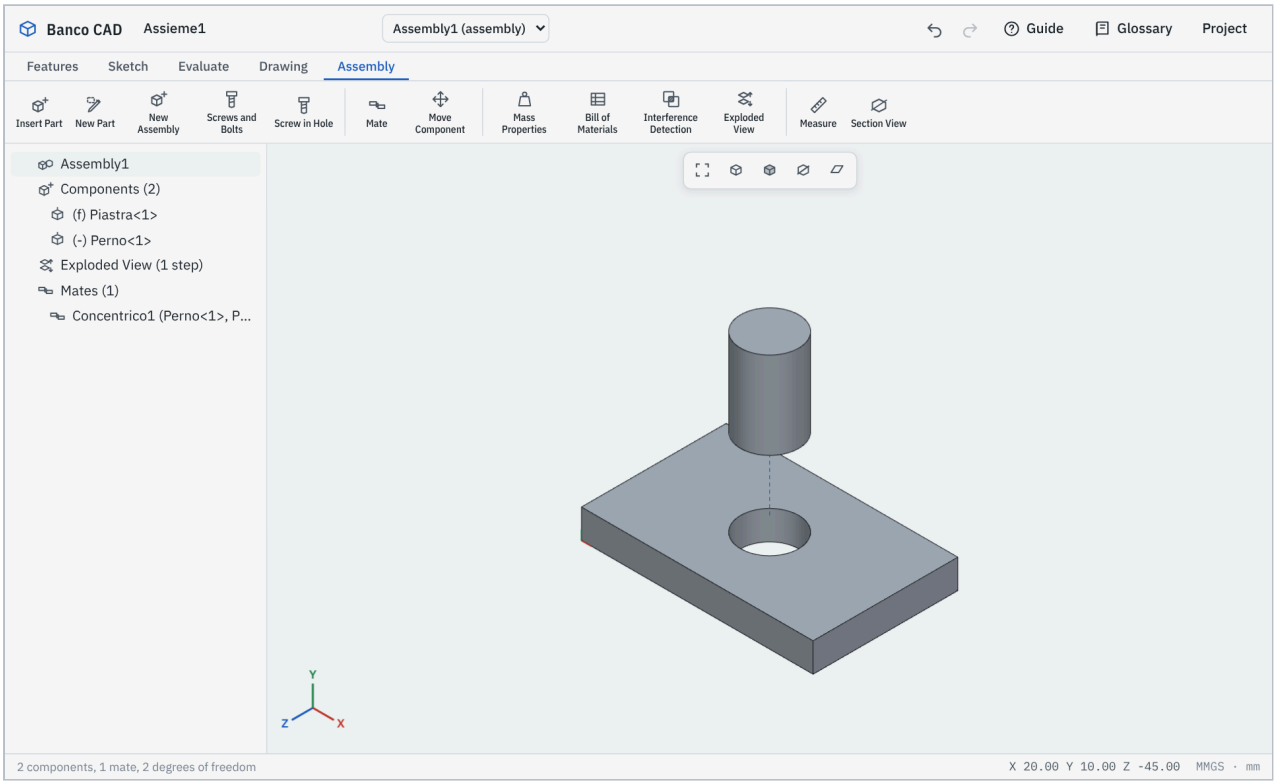
4. To be quicker: **Arrange automatically** puts the components in a row along the chosen direction, with the spacing you give.
5. In the tree, a click on **Exploded View** explodes or collapses it; a double-click edits the steps.

> **Tip** The real positions do not change: mass and interferences are calculated on the assembled machine.

> **Tip** In the exploded view the mouse turns the view. Mate and Move Component collapse the view by themselves.

> **Tip** For a slanted direction press **Direction from a face** and click a face: flat, it goes along the normal; cylindrical or conical, along the axis.

> **Tip** Export STEP and Save PDF work on what you see.



Exploded view: the pin lifted 40 mm out of the hole.

9.12 Making the assembly drawing

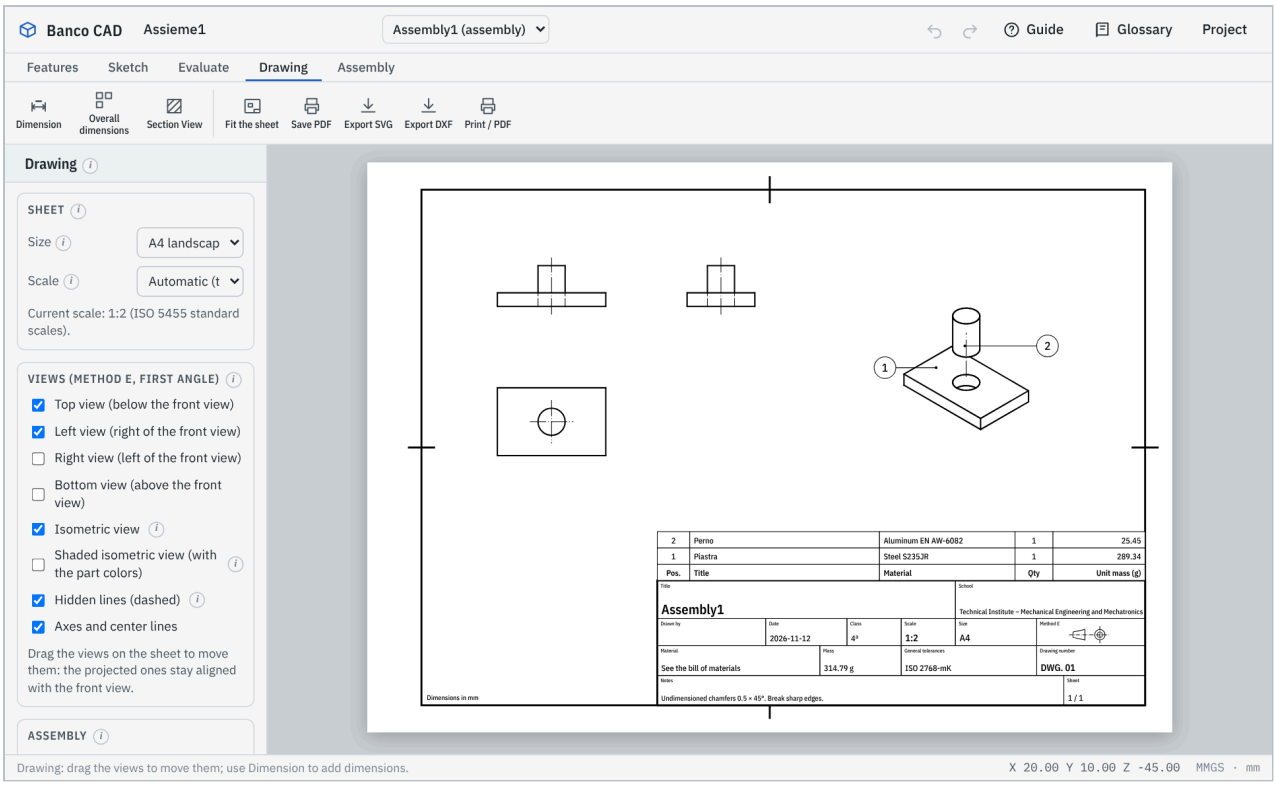
Get the general drawing to hand in: views, numbered pieces and bill of materials.

1. While in the assembly, open the **Drawing** tab: the assembly drawing opens (from a part, the part's drawing opens).
2. In the panel, **Assembly** group: **Bill of Materials**, **Balloons** and, if there is an exploded view, **Exploded isometric view**.
3. Dimensions, section, title block, PDF, SVG and DXF work as in the part drawing.

> **Tip** Each balloon number is the same as its row in the bill of materials. Row 1 sits right above the header, as the ISO 7573 standard requires.

> **Tip** The balloons place themselves along the sides of the isometric view. To move one, drag it with the mouse: the line that links it to the part follows it. Choose the scale with the isometric view already in place.

> **Tip** In the section all the components have the same hatch.



Assembly drawing with bill of materials and balloons.

9.13 Checking whether components overlap

Find out before building that two pieces take up the same space, and by how much.

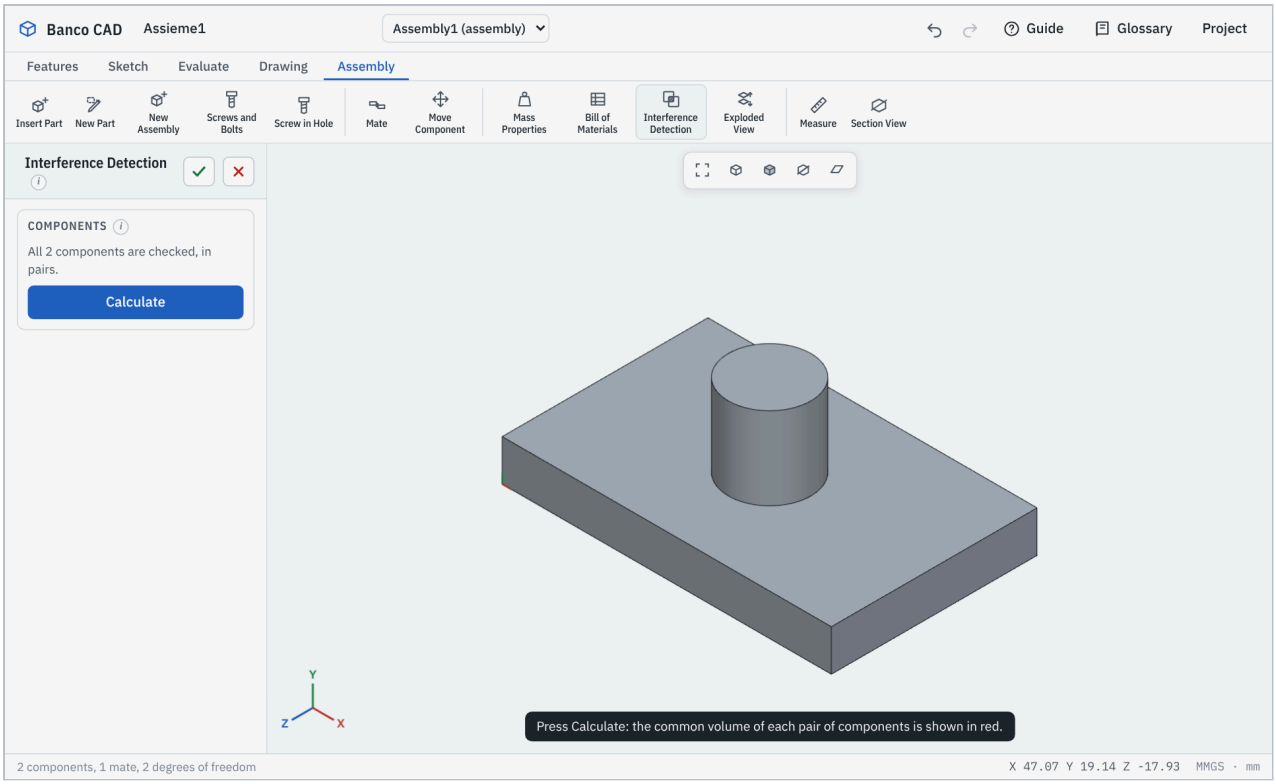
Where to find it: Assembly tab › Interference Detection

1. Put the components in the position to check.
2. Press **Interference Detection**, then **Calculate**.
3. In the list each row gives the two components and the shared volume. Click a row: that volume is shown in red, even inside the pieces.
4. Fix the part or the mate and calculate again.

> **Tip** Two faces in contact are not an interference: a pin with the same diameter as the hole is not reported.

> **Tip** With imported pieces (STEP, IGES, STL, OBJ) a very thin layer is marked «maybe just contact», because those pieces are made of triangles.

> **Tip** With very complex pieces the calculation can hold the page still for a few minutes.



The Interference Detection panel.

10 The drawing

The drawing is the technical drawing of the part or of the assembly, to ISO standards, ready to print or hand in.

10.1 Making the part drawing

The dimensioned technical drawing to hand in or print.

Where to find it: Drawing tab › Overall dimensions

1. Open the **Drawing** tab: the views appear in Method E.
2. On the left choose the Sheet (A4, A3…) and the Scale (Automatic chooses the largest standard scale that fits on the sheet).
3. Check the views you need (from above, from the left… and the isometric view) and the Hidden lines.
4. If you want the isometric view in color, check Shaded isometric view: the faces take the color of the part (also that of a STEP file), the other views stay as lines.
5. Drag the views to arrange them: the projected ones stay aligned with the front view.
6. Fill in the Title block (title, author, class, drawing number).
7. **Fit the sheet** brings the whole drawing into the window; the wheel zooms the sheet in or out around the pointer.

10.2 Dimensioning the drawing

Put the sizes on according to the rules of technical drawing.

Where to find it: Drawing tab › Dimension

1. **Overall dimensions:** adds in one go the largest sizes and the diameters of the circles.
2. **Dimension:** click an edge (length), a circle (diameter) or two points of the same view, then click where to place the dimension.
3. Dimensions can be dragged to arrange them; the «Dimensions» list on the left lets you delete them.

> **Tip** Dimensions go on the orthographic views, not on the isometric view.

10.3 Section view on the drawing

Show the inside of the part with hatching.

Where to find it: Drawing tab › Section View

1. Press **Section View** (Drawing tab).
2. Choose the view to section, the position of the cutting plane and the letter (A).
3. «Apply»: the view becomes the hatched section A-A and the trace of the plane appears in the nearby view.

10.4 Printing and exporting the drawing

Hand in the drawing on paper or as a file.

Where to find it: Drawing tab › Print / PDF

1. **Save PDF:** the simplest way: a PDF file of the drawing, to scale, that opens and prints anywhere (see «Saving the drawing, the sketch or the 3D view as PDF»).
2. **Print / PDF:** in the print window choose «Actual size» or «100 %» to get the right scale, or «Save as PDF».
3. **Export SVG:** vector drawing at 1:1 scale.
4. **Export DXF:** for AutoCAD and other 2D CAD programs, with layers and line types.

10.5 Saving the drawing, the sketch or the 3D view as PDF

Have a file to hand in, send or print, that opens on any computer, tablet or phone.

Where to find it: Drawing tab › Save PDF

1. **Drawing:** on the Drawing tab press **Save PDF**. The PDF has the real sheet size (A4, A3, A2) and the drawing at 1:1 scale: lines, dimensions and title block stay sharp at any zoom.
2. **Sketch:** with the sketch open choose **Project › Save PDF of the sketch**. The sketch goes on a landscape A4, at the largest standard scale that makes it fit (for example 2:1), written at the bottom. Dimensions and colors are those of the screen.
3. **3D view:** outside the sketch and the drawing choose **Project › Save PDF of the 3D view**. The view goes in as it is on screen (orientation, style, zoom) on a landscape A4, on a white background.
4. In the shared page the file goes through the claude.ai save window; in the file opened from the computer it goes into Downloads.

> **Tip** To print to scale, open the PDF and in the print window choose «Actual size» or «100 %» (not «Fit to page»).

> **Tip** The 3D view in the PDF is an image: for a sharp drawing to scale use the drawing.

10.6 Opening the part in another CAD program (Export STEP)

Continue the part in SolidWorks, Inventor, Fusion or FreeCAD, or send it to whoever has to make it.

1. Choose **Project › Export STEP (for other CAD programs)**: the .step file holds the part as a solid, in millimeters.
2. In the other program open it with **File › Open** (in SolidWorks: STEP file type).
3. Planes, cylinders, cones, spheres and tori arrive as true surfaces: in the other CAD program a hole has its exact diameter and a fillet its radius. The message says how many there are.
4. Surfaces that cannot be written exactly (two cylinders crossing each other, a sweep along a free curve) arrive divided into narrow flat faces: the solid stays closed.

> **Tip** Dimensions and the feature history do not pass: no CAD program exchanges them with the others. New features can be added on top of the imported part.

> **Tip** Banco CAD reads its own STEP files back with Project › Import STEP or IGES.

11 Files: what opens and what is written

11.1 Opening files from other programs

Format	Command	What it becomes	Good to know
STEP (.step, .stp), **IGES** (.igs, .iges)	Project › Import STEP or IGES	"Imported" solid, with the true faces divided into fine triangles	Units read from the file, colors of the file, choice of bodies. At most 100,000 triangles: beyond that, the surfaces are divided more coarsely.
STL, **OBJ**	Project › Import STL or OBJ	Faceted "Imported" solid	You choose the units (mm, cm, m, inches). At most 100,000 triangles. Warning if the body is not closed.
DXF, **DWG**	Project › Import DXF or DWG (sketch); or Insert DXF or DWG in the Sketch tab	Lines, arcs and circles in a sketch	Choice of layers, preview, no limit on entities. DWG from R1.4 to 2018. Texts, dimensions and hatches become a reference background.
Project (.json)	Project › Import a project	The project as it was	The file is the one from Export project.

> **Remember** Imported parts arrive **without the feature history**: the original dimensions cannot be edited, but new features can be added on top. Sketches from DXF arrive without dimensions, so blue: you can dimension them.

The proprietary files of SolidWorks (.sldprt, .sldasm, .slddrw) and of Inventor (.ipt, .iam, .idw, .ipn) **cannot be opened**, because their specifications are not public. From the original program, export parts and assemblies to STEP, drawings to DXF.

11.2 Saving for other programs

Format	Command	Good to know
Project (.json)	Project › Export project	All the work, to open again in Banco CAD.
PDF	Save PDF	The drawing is vector, on the real sheet, at 1:1 scale. The sketch goes on a landscape A4 with the largest standard scale that fits it. The 3D view is an image.
STEP (AP214)	Project › Export STEP	In mm, with the true surfaces (planes, cylinders, cones, spheres, tori). One solid per body or per component.
STL	Project › Export STL	For 3D printing.
SVG, **DXF**	Drawing tab	The drawing. In the DXF the isometric view stays as lines, even if it is shaded on screen.

11.3 The DWG → DXF Converter

Banco CAD opens DWG files directly. The **DWG → DXF Converter** is a separate program, for those who want to save the DXF files, use them in other programs or convert many drawings at once: **rayuk-creator.github.io/CAD/convertitore-dwg**.

12 Common problems and limits

12.1 If something goes wrong

What happens	What to do
The program “does not load”, the page looks half done	Check the address: the right one is **rayuk-creator.github.io/CAD**, not the repository page on github.com.
I cannot find my work	Your work is in the browser where you made it. Website and downloaded file are separate, and so are two browsers or two computers. To move it use Export and Import project.
The sketch is blue	Dimensions or relations are missing. You can extrude it anyway, but it is better to finish it: a part that is all black does not change by mistake.
The sketch is red	There are too many dimensions or relations, or they conflict. Remove the last one you added, or use Undo.
The profile does not extrude	The outline must be closed and must not cross itself. The red message in the panel says what is wrong.
An exclamation mark “!” in the assembly tree	An element of the mate has disappeared, for example after a change to the part, or the mate is impossible. Open it and choose the elements again.
A .sldprt or .ipt file does not open	They are proprietary formats: export to STEP from the original program.
A DWG inside a .zip does not open	The message says so: extract the DWG from the zip first.
I want to know if the calculations are right	**Evaluate › Self-check** tab: the program runs its tests again with values calculated by hand.

12.2 Limits to know

- **Texts in the PDF:** they use Helvetica; characters outside the Western alphabet become “?” (for example ⌀ becomes Ø).
- **PDF of the 3D view:** it is an image at screen resolution. For a sharp print use the PDF of the drawing.
- **Shaded isometric view:** with very intertwined parts some face may cover another one the wrong way.
- **Drawing background:** tables in DWG files from 2010 on cannot be read; the width of texts may differ a little from AutoCAD.
- **Screws and threads:** the thread is cosmetic, except with the helix option; threads are not drawn in the isometric view; lengths follow the ISO 888 series, but the ranges available for each size depend on the supplier.
- **Interferences:** a single pair of huge parts can freeze the page for a minute or more.
- **Assemblies:** a subassembly inside a flexible subassembly stays rigid; the free-form surfaces of a STEP file can only be mated with Tangent, against a plane or a sphere.
- **Auto-select of the scope:** it is not recalculated with the panel closed. For bodies added later, open the feature again.

- **Loft:** no guide curves and no tangency imposed at the ends; each profile is a single closed outline, on a plane of its own.
- **Helix and spiral:** the profile always turns with the curve (“Follow path”) and must be drawn on a plane that goes through the start of the curve. The curved faces of springs and lofts are facets: in the STEP file they stay flat.

13 Glossary

All the terms of the program, in alphabetical order. In brackets the tab or the subject they belong to.

#

118° drill point (Holes) The conical bottom left by the twist drill in blind holes. The depth of the hole is measured up to the start of the cone.

3 Point Arc (Sketch) Arc passing through three points: start, end and a point in between.

A

About and licenses (Interface) Project › About and licenses: who wrote the program, its license, the parts by other authors and the trademarks mentioned.

Add Relation (Sketch) Geometric constraints between entities (horizontal, parallel, tangent...). Together with dimensions they make the sketch fully defined. Select the entities (Ctrl+click) and choose the relation.

Angle (Features) For the revolve: how much the profile rotates (360° = full turn). For the circular pattern: the angle over which the instances are spread.

Angle mate (Assembly) Two faces or two edges at a given angle: the angle is measured between the outward normals of the faces and between the directions of the edges. Between an edge and a face, 0° means the edge is parallel to the face.

Assembly (Assembly) Several parts mounted together, each in its place. In Banco CAD the assembly and its parts are all in the same project, so in a single file.

Assembly degrees of freedom (Assembly) The movements a component still has: at most 6 (3 translations and 3 rotations). In the tree (-) means it still has some; without a sign it is defined, that is it no longer moves.

Assembly drawing (Assembly) The drawing of the whole assembly: views, isometric view (exploded too), balloons and bill of materials above the title block. It opens with the Drawing tab when you are in the assembly.

Assembly mass properties (Assembly) Mass, volume, area, center of mass and overall size of all the components together. Each part uses its own material; the center of mass is weighted by mass. The volume shared by two interpenetrating components counts twice.

At the center (Relations) The point coincides with the center of a circle or an arc.

Auto-select (Features) Inside "Feature Scope", already checked in new features and when you choose "Selected bodies" (as in SolidWorks): it checks by itself the bodies the feature touches; clear it to choose them by hand. The list is updated while the panel is open, then it stays saved in the feature: if you add other bodies later, reopen the feature or use "All bodies" to include them.

Axis (References) Reference line: for circular patterns it is the axis of rotation; it can be taken from one of the X, Y, Z axes or from the part (a cylindrical face or a circular edge).

Axis of Revolution (Features) The centerline of the sketch around which the profile rotates.

B

B-spline (Import) A smooth curve defined by a few control points. In the STEP file it describes the edges that are not lines, circles or ellipses, for example the edge between two cylinders of different diameter.

Balloon (Assembly) The number in a circle that marks a piece in the assembly drawing: it is the same number as its row in the bill of materials.

Biarc (Import) Pair of arcs tangent to each other, which starts and ends with the same direction as the curve to imitate. With many biarcs one after the other an ellipse or a spline is followed without corners: each biarc goes as far as possible along the curve, with the joining point of the two arcs chosen to stay closer to the curve, as long as the deviation stays below 1/10,000 of the smallest dimension of the curve (a 100 × 60 ellipse becomes about 36 arcs). The spline always passes exactly through its fit points. In the part the faces of these arcs look like a single surface, without lines between them, and their centers do not appear in the sketch.

Bill of Materials (Assembly) The list of the parts needed to build the assembly: number, name, material, quantity and mass. Components of the same part make a single row.

Blind (Options) The feature stops at the depth typed. A blind hole does not come out on the other side.

Bodies of an assembly (Import) A STEP file can contain several separate solids, for example the pieces of an assembly. After reading, Banco CAD lists them with the number of faces and the volume: with the check boxes you choose which ones to import. With the check box "One «Imported» for each body" (checked) each body arrives in its own feature, with the name of the body if the file has it; without it, they arrive together in a single "Imported". In an IGES file the surfaces that touch form a single body.

Bodies to Pattern (Features) In the linear pattern, the circular pattern and the mirror: repeats whole bodies instead of single features. Each copy is a new body, with its holes and cosmetic threads too.

Bounding box (Evaluate) The maximum dimensions of the part along X, Y and Z: the smallest box that contains it.

C

Center lines (Drawing) The thin dash-dot lines that mark the axes of holes and cylindrical parts (ISO 128).

Center of mass (Evaluate) The point where all the mass can be thought to be concentrated (X, Y, Z with respect to the origin).

Center Rectangle (Sketch) Rectangle given by the center and a corner: it is symmetric about the center, handy for parts centered on the origin.

Centerline (Sketch) Dash-dot construction line: it creates no material. It is used as the axis of a revolve, as the symmetry axis for "Mirror Entities" and for the Symmetric relation.

Centerpoint Arc (Sketch) Arc given by the center, the start point and the end point.

Chamfer (Features) Cuts an edge with an inclined face. Types: equal distance (45°), distance and angle, two distances. On shafts it is specified for example as 1×45°.

Circle (Sketch) Circle given by the center and a point on the circumference. It is dimensioned with the diameter (Ø).

Circular Pattern (Features) Repeats features around an axis, for example the holes of a flange. With "equal spacing" the instances are spread over the total angle.

Clearance hole (Holes) Hole slightly larger than the screw, so it passes without forcing. The ISO 273 standard gives three series: close (fine), normal (medium), loose (coarse). For M6: Ø6.4 / Ø6.6 / Ø7.

Clockwise (Features) The direction in which the helix turns, looking at the sketch from the front. Counterclockwise while rising is a right-hand helix, like ordinary screws and bolts; clockwise is a left-hand helix. "Reverse direction" makes the helix rise on the other side of the plane, with the same sense of rotation.

Closed body (Import) A faceted body is closed when every triangle side is shared with exactly one other triangle: then it encloses a volume. If there are free sides (holes in the surface) volume and mass are not reliable and the features added on it may fail.

Coincident (Relations) A point lies on another point, on a line or on an arc. It is the relation that closes contours.

Coincident mate (Assembly) Two entities in the same place: two faces on the same plane, a vertex on a face, two edges on the same line, two vertices at the same point. Two circular edges: same center and same plane.

Collinear (Relations) Two lines lie on the same straight line.

Combine (Features) Works on bodies that already exist. Add merges them into one; Subtract removes the material of the others from the main body; Common keeps only the material they all share. The bodies used disappear and one remains.

CommandManager (Interface) The bar at the top with the Features, Sketch, Evaluate, Drawing and Assembly tabs. Each tab shows the commands of that group, as in SolidWorks.

Component (Assembly) A part placed in the assembly, with its position. The same part can appear several times: Screw<1>, Screw<2>. They are copies of the same part: if you modify the part, they all change.

Concentric (Relations) Circles or arcs with the same center.

Concentric mate (Assembly) Two round entities on the same axis: cylinders, cones, circular edges. Between two spheres: same center. With a vertex: the vertex lies on the axis.

Confirmation Corner (Interface) The ✓ and ✕ buttons at the top right of the view: they confirm or cancel the open feature or sketch, like the PropertyManager.

Construction Geometry (Sketch) Dashed entities that help you draw (axes, reference circles) but do not become faces of the solid. The "Construction Geometry" command converts the selected entities and back.

Convert Entities (Sketch) Copies the selected edges of the part (in gray) into the sketch, linked to the model: if the part changes, they change too.

Corner Rectangle (Sketch) Rectangle given by two opposite corners. The horizontal and vertical relations are added by themselves.

Cosmetic thread (Assembly) The thread is not a true helix: the part contains the cylinder at the nominal diameter (screw) or the tap drill hole (tapped hole, nut), and the thread is drawn with thin lines. On the drawing, as ISO 6410 requires: thin line on the minor diameter of the screw, $d_3 = d - 1.22687\ P$, or on the nominal diameter of the hole; seen end-on, 3/4 of a thin circle. This keeps the model light. The mass of the screw is a little higher than the real one.

Counterbore (Holes) Cylindrical enlargement at the top of the hole that takes the head of a socket head screw (ISO 4762), so the head does not stick out. Sizes from DIN 974-1.

Countersink (Holes) 90° conical enlargement at the top of the hole for countersunk screws (ISO 10642): the head stays flush with the face. Sizes from ISO 15065.

Curve deviation (Import) The largest distance between the true curve (ellipse or spline of the DXF file) and the arcs that replace it. The import message says how much it is, in millimeters: usually a few thousandths, so volume and mass stay exact well within 1%.

Curved edge (Features) The boundary of a face made from an ellipse or a spline imported from a DXF file. In the part it is made of many tangent arcs, but as in SolidWorks it counts as a single edge: one click selects all of it (it is colored entirely) and the fillet or chamfer follows it with no lines between one piece and the next. The face of the curve is also selected all together.

D

Defined by (Features) Which two measurements describe the helix, as in SolidWorks: pitch and revolutions, height and revolutions, or height and pitch; the third one is worked out by itself (height = pitch × revolutions). "Spiral" makes a flat curve: the pitch is how much the radius grows at each turn.

Delete/Keep Body (Features) Removes the chosen bodies, or keeps only those. Like any feature it can be suppressed or deleted: the bodies come back.

Depth (Options) How far the extrusion or cut extends from the sketch plane, in millimeters.

Direction 1 (Options) The main direction of the feature, normal to the sketch plane.

Direction 2 (Options) Extends the feature on the other side of the sketch plane too, with its own depth.

Distance (Options) For planes: the distance from the reference. For patterns: the spacing between one instance and the next.

Distance mate (Assembly) Two entities at a given distance, in millimeters. Two faces also stay parallel. For two faces, Flip side chooses on which side the other entity lies.

Draft (Options) Tilts the walls of the extrusion by the angle typed: the profile narrows as you move away from the sketch plane (or widens with "outward"). It is used for molded or cast parts, so they come out of the mold. Flat faces become inclined planes, cylindrical ones cones.

Drawing (Drawing) The 2D technical drawing obtained from the model: orthographic projection views, dimensions, title block. If the part changes, the drawing updates.

Drawing background (Import) The texts, dimensions, leaders, hatches, geometric tolerances and tables of a DXF or DWG file, shown in gray under the sketch, only to look at: they are used to read measurements and notes while you dimension. They are not entities: they cannot be selected, you cannot snap to them, they do not change the contours or the state of the sketch. The dimensions of the file do not drive the shape: the measurement must be set again with Smart Dimension. It is shown only while the sketch is being edited, not in the PDF, the 3D part or the STEP file. In the sketch panel: Show the background check box and Remove the background button. Texts smaller than 3 pixels on the screen are not drawn: they appear when you zoom in. The texts use the program's font, so their width may differ a little from AutoCAD's.

Drawing hatch (Import) An area of the drawing filled with a pattern of lines or with a solid color (HATCH in the DXF or DWG file). In the background it is shown in light gray, with the pattern and the holes of the file. A pattern that is too dense (more than 30,000 strokes, or 200,000 in the whole drawing) and solid fills are shown as a light fill: a pale color, without lines.

Drawing table (Import) An AutoCAD table (ACAD_TABLE), for example a parts list. In the background it is shown as in the DXF file and in DWG files up to 2007. In DWG files saved from 2010 onward the LibreDWG reader does not give the position of the table: it is left out, counted among the skipped items in the message. To get it, save the drawing in AutoCAD as DXF.

Driven dimension (Sketch) Information-only dimension, in parentheses: it shows a measurement but does not impose it. You get it when a normal dimension would over-define the sketch.

DWG → DXF Converter (Import) Separate free page that turns AutoCAD drawings (.dwg) into DXF files to save: it is used to pass them to other programs or to convert many at once. To open a DWG here it is not needed: you choose it directly (see DWG file). It uses the same free reader, LibreDWG, and works in the browser, without sending the file anywhere. It shows the preview of each drawing with its overall size (what Banco CAD will read) and can save all the DXF files in a single .zip, from which Banco CAD lets you choose the drawing. It starts with little memory (16 MB) and takes more only for large drawings, so it also works on tablets.

DWG file (Import) The format of AutoCAD drawings, from version R1.4 to 2018 (the newer ones still use the 2018 format; if it changes, it is tried anyway). It opens directly: the free LibreDWG reader turns it into a DXF inside the page, without sending the file anywhere, then the drawing follows the same path as a DXF file, with preview and layers. Texts and dimensions can arrive as Drawing background; hatches are skipped, and the final message says how many there are. Splines arrive with only their fit points: the curve rebuilt here can differ from AutoCAD's by a few tenths of a millimeter per meter (in the tests 0.1 and 0.3 mm/m). The first reading takes a moment, meanwhile the page stays free.

DXF file (Import) Exchange format for 2D drawings, born with AutoCAD and read by almost all CAD programs. Here text (ASCII) DXF files are read: lines, arcs, circles, polylines (with curved segments too), ellipses, splines and blocks. Ellipses and splines become arcs tangent to each other (biarcs). Texts, dimensions and hatches are skipped. You can also choose a .zip with the DXF inside. See also DWG file.

E

Edge (References) The line where two faces meet. Fillets and chamfers are made on edges; a straight edge can also give the direction of a linear pattern.

Edges to Fillet (Features) The edges the fillet or chamfer is applied to: click the edges in the view (they turn blue); a second click removes them.

Ellipse (Import) "Flattened" closed curve, like a circle seen at an angle: it has a major semi-axis a and a minor one b; the area is π·a·b. In DXF files it can be whole or only a part (elliptical arc). In Banco CAD, as in many CAD programs, the imported ellipse becomes a chain of tangent arcs (biarcs) very close to the true curve.

End Condition (Options) Tells where the extrusion or cut ends: Blind (at a depth), Through All (crosses the whole part), Mid Plane (half the depth on each side).

Equal (Relations) Lines of the same length, or circles and arcs of the same radius. So one dimension is enough for all of them.

Exit Sketch (Sketch) Closes the sketch and saves it in the tree. You can also press a feature command directly (Extruded Boss/Base...): the sketch is confirmed by itself and used.

Exploded View (Assembly) The components move apart from each other to show how they are assembled. It is made of steps: which components, in which direction, how far. The true positions and the mates do not change: when you collapse it everything goes back in place.

Export DXF (Drawing) Exchange format for AutoCAD and the other 2D CAD programs, with ISO layers and line types.

Export STEP (Import) Project › Export STEP saves the part in a .step file that opens as a solid in SolidWorks, Inventor, Fusion, FreeCAD and the other 3D CAD programs, in millimeters. Planes, cylinders, cones, spheres and tori (holes, fillets, conical chamfers, spherical corners, revolves) arrive as true surfaces, with circle or ellipse edges: the other CAD program sees a real hole, with its diameter. Where two different surfaces cross (two cylinders of different diameter) the edge is a B-spline: there too the surfaces stay true.

Export STL (Drawing) The format for 3D printing: the solid described by many triangles.

Export SVG (Drawing) Vector drawing at 1:1 scale on paper: it opens in the browser and in graphics programs.

External geometry (Sketch) The edges of the face you are drawing on, shown in gray. You can snap and dimension to them (for example the distance of a hole from the edge).

Extruded Boss/Base (Features) Takes a closed contour of the sketch and "pulls" it perpendicular to the plane, creating material: a rectangle becomes a box, a circle a cylinder.

Extruded Cut (Features) Like the extrusion, but it removes material: pockets, shaped holes, slots.

F

Face (References) A surface of the solid bounded by edges. Flat faces can be used for sketches, planes and holes; cylindrical ones give the axis for circular patterns.

Faces of the original model (Import) The faces of the part as the original program created them: a flat face, the wall of a hole, a fillet. In STEP and IGES files Banco CAD knows them: the edge is drawn only where one face ends and another begins, even between two tangent faces (like the tangent edges of SolidWorks), while the seam of a cylinder is not shown.

Faceted body (Import) Solid described by many flat triangles instead of exact surfaces: a cylinder becomes a prism with many sides. The more triangles there are, the more faithful the shape. Where neighboring facets differ by less than 30° the edge is not drawn, so curved surfaces look smooth.

Fastener chamfers (Assembly) The tip of the screw is chamfered at 45° down to the minor diameter; the hexagon head and the nut have the 30° chamfer starting from a circle of 0.95 times the width across flats. It is the usual drawing, not a value guaranteed by the standard for every manufacturer. It is removed with the check box in the screws window.

Feature (Features) Every operation that creates or modifies the solid (extrusion, hole, fillet...). The features are in the tree and can be edited at any time: the part updates by itself (parametric modeling).

Feature Scope (Features) In a part with several solid bodies: which bodies a feature works on (cut, hole, extrusion, revolve, sweep, rib, imported body). A new feature starts from "Selected bodies" with "Auto-select" checked, as in SolidWorks. If you clear the check box you check the bodies in the panel, click them in the view or click their name in the tree above the view, and the others are not touched. "All bodies" works on all of them, including those added later. A feature that adds material merges only with the chosen bodies it touches: if it touches none, it becomes a new body, with a warning. With "Merge result" cleared the scope is not needed and is not shown. The patterns and mirrors of that feature work on the same bodies if you chose them by hand; with Auto-select each copy works on the bodies it touches.

FeatureManager design tree (Interface) The list on the left with the planes, the origin and all the features in the order they were created. The part is rebuilt following the tree from top to bottom: double-click a feature to edit it, right-click for the other commands.

Features to Pattern (Features) The features (extrusions, cuts, holes, sweeps, ribs...) that the pattern or the mirror copies. They are chosen with a click on one of their faces in the view or on the name in the tree.

File colors (Import) The colors that the original program wrote in the STEP or IGES file (of the part or of single faces). Banco CAD shows them on the faces of the imported body; they are removed with the "File colors" check box (double-click "Imported"). They change neither the material nor the mass.

File units (Import) The unit in which the coordinates of the imported file are written (millimeters, centimeters, meters, inches). STL and OBJ files do not state it: if the part arrives 10 times smaller or larger than expected, the chosen unit is wrong. In STEP and IGES files, instead, the unit is written in the file and is converted to millimeters by itself.

File Z axis up (Import) In Banco CAD (as in SolidWorks) the vertical is the Y axis; many 3D printing programs use Z instead. With this check box the Z of the file becomes the vertical and the part does not arrive lying down.

Fillet (Features) Rounds the selected edges with a circular surface of given radius. On an outside (convex) edge it removes material, on an inside (concave) one it adds it. All the edges of the same feature are filleted together, as in SolidWorks: where two fillets meet they blend with each other, where three meet at right angles the corner becomes a portion of a sphere. You can fillet straight, circular and arc edges and the boundary of an imported ellipse or spline (curved edge), with a radius smaller than that of the curve.

Fix (Relations) The entity can no longer move. Use it sparingly: it is better to define the sketch with dimensions and relations to the origin.

Fixed part (Assembly) A component that does not move: the others are placed with respect to it. In the tree it has (f) in front of the name. The first part inserted is fixed; right-click to choose Fix or Float.

Flexible subassembly (Assembly) A subassembly whose pieces move also in the assembly that contains it, according to the mates they have inside. It is chosen with a right-click on the component: Make flexible. Each copy has its own positions and the original assembly does not change. Make rigid puts the pieces back where they are in their own assembly. A subassembly inside a flexible one stays rigid.

Flip alignment (Assembly) Chooses between the two solutions of the same mate: two faces facing each other or facing the same way, a pin turned one way or the other.

Front Plane (References) Vertical standard plane containing the X and Y axes (you look at it from the front). It is used for the main profile of parts seen from the front.

Fully Defined (Sketch) The sketch is black: shape and position are fixed by relations and dimensions, no point can be dragged any more. It is the right state for a good model (in SolidWorks: "Fully Defined").

G

General tolerances (Drawing) The tolerances that apply to all dimensions without a written tolerance. ISO 2768-mK: medium class for sizes, class K for form and position.

Geometric tolerance (Import) The frame that gives the form or position tolerance (ISO 1101): symbol, value, datums. The file gives only the content, so in the background the frame is rebuilt, 2 times as tall as the text: position and content are those of the file, the widths may differ a little from AutoCAD. The symbols are drawn with strokes.

GNU GPL 3 (Interface) The license of Banco CAD. It lets anyone use, study, modify and redistribute the program, as long as the distributed versions keep the same license and the source code is available.

Guide (Interface) The panel with step-by-step instructions for every command: what it is for, how to do it, tips and terms. The "Show me where" button opens the right tab and makes the command blink.

H

Helix and Spiral (Features) Curve in space that starts from a circle drawn in a sketch: the helix turns around the axis of the circle and rises at the same time (springs, threads); the spiral stays on the plane and widens at each turn (spiral springs). It creates no material: it is used as the path of a Sweep or a Swept Cut. In SolidWorks: Insert › Curve › "Helix/Spiral".

Hidden lines (Drawing) The edges that cannot be seen because they are covered by the part, drawn with a dashed line (ISO 128).

Hole type (Holes) Simple (free diameter), clearance for screw (ISO 273), counterbore for socket head screws, countersink for countersunk screws, tapped (tap drill hole for tapping).

Hole Wizard (Features) Holes according to the standards: you choose the type and the screw (M3...M20) and the diameters are taken from the tables. You click a flat face and place the positions of the centers.

Horizontal (Relations) The line (or the line joining two points) is parallel to the X axis of the sketch.

I

IGES (Import) Exchange format older than STEP (born in the 1980s): it describes the surfaces of the part, often one by one without saying how they are joined. Banco CAD stitches them together; if some gap remains, the body-not-closed warning appears. When possible, STEP is better.

Import options (Import) The choices needed to read the file correctly: the unit of measurement and which axis points up. They can be changed at any time by double-clicking "Imported" in the tree.

Import STEP or IGES (Import) Brings into Banco CAD a part, or a small assembly, made with another CAD program (Project › Import STEP or IGES). The file is read from its true surfaces; the "Imported" feature appears in the tree: the part can be seen, measured and put on a drawing, its volume and mass are calculated, and sketches and features can be added on it. Curved surfaces are divided into very dense triangles: the error on volume, area and mass stays well below 1%. The unit of measurement is read from the file.

Import STL or OBJ (Import) Brings into Banco CAD a part made with another program (Project › Import STL or OBJ). The "Imported" feature appears in the tree: the part can be seen, measured and put on a drawing, and its volume and mass are calculated. The original dimensions cannot be edited, but you can draw sketches and add features on it (cuts, holes, extrusions).

Imported file (Import) The name and format of the file the body comes from, with the number of triangles (for STEP and IGES: of faces of the original model), the flat faces recognized and the overall size. The flat faces (made of several triangles on the same plane) can be used for sketches, like the faces of a part drawn here.

Insert DXF or DWG (Import) Brings into a sketch the lines, arcs and circles of a 2D drawing made with another program (AutoCAD, DraftSight, LibreCAD, the SolidWorks drawing...). Outside a sketch, or from Project › Import DXF or DWG (sketch), it creates a new sketch on the chosen plane; inside a sketch it adds the geometry to the open one. The entities arrive without dimensions, so blue (under defined): they can be extruded or revolved straight away, or dimensioned. Texts and dimensions of the file can arrive as Drawing background.

Insert Part (Assembly) Adds a component to the assembly: a part of the project, or a STEP, IGES, STL, OBJ file or an exported project, which is copied into the project.

Instances (Features) The total number of copies in a pattern, counting the original too.

Interference (Assembly) Two components occupying the same space: in reality they could not be assembled like that (unless it is intended, like a press fit). It is measured by the shared volume. Two faces that only touch are not an interference.

Interference Detection (Assembly) The command that checks all the components in pairs and shows the shared volume of each pair, also in red in the view. With parts made in the program the volume is calculated on the true surfaces; with imported ones, made of triangles, a very thin layer is marked "maybe only contact".

Isometric (Interface) View in which the three axes X, Y, Z form equal angles on the screen: three faces of the part are seen together. It is the standard view for the 3D model.

L

Language (Interface) The program is in Italian; with Project › Language: English it becomes English, with the command names of the English version of SolidWorks. To go back, choose Project › Lingua: italiano. The project stays the same; in English numbers use the decimal point.

Layer (Import) In a DXF (or DWG) drawing each entity lies on a "layer", as on a transparent sheet: for example PART, DIMENSIONS, TITLEBLOCK. In the DXF import window you choose with the check boxes which layers to import; those turned off or frozen in the file start unchecked. The entities of a block lying on layer "0" take the layer of the inserted block, as in AutoCAD.

Line (Sketch) Consecutive segments: one click for each vertex, double-click or Esc to finish. With numeric input you can type length and angle.

Line type (Import) The way a line is drawn: continuous, dashed (hidden edges) or dash-dot (axes). In the DXF import, dash-dot lines become centerlines, dashed ones construction geometry; continuous ones form the contours.

Linear Pattern (Features) Repeats one or more features (holes, cuts, bosses...) along one or two directions, with spacing and number of instances.

Loft (Features) Creates a solid that passes through two or more closed profiles, drawn on different planes: twisted blades, hoppers, transitions between a square and a round section. Each profile is a sketch with a single contour; the order in which you choose them is the order in which the solid passes through them. Between two profiles the faces are straight (ruled); with three or more the solid passes through all of them with a smooth curve. In SolidWorks: "Lofted Boss/Base" (here without guide curves).

Lofted Cut (Features) Like the Loft, but it removes material: pockets that change shape from the opening to the bottom, ducts that narrow.

M

Mass (Evaluate) Volume × density of the material. With the material "Not specified" 1 g/cm³ is used.

Mass Properties (Evaluate) Volume, surface area, mass and center of mass of the part, calculated with the exact surfaces (cylinders, cones, spheres, tori), as in SolidWorks.

Mate (Assembly) A link between two entities of two components (faces, edges or vertices). Each mate removes degrees of freedom: the solver moves the free components as little as possible to satisfy them all.

Mate type (Assembly) The way the two chosen entities are linked: coincident, concentric, parallel, perpendicular, distance or angle. Only the types possible with those entities are active.

Mate vertex (Assembly) A vertex chosen in a mate is remembered together with the faces that meet there. If the dimensions of the part change, the program looks for the vertex with those faces, the one closest to where it was. If that vertex no longer exists, for example after a chamfer right there, the mate gets the exclamation mark. In an imported body the position counts, because the body does not change.

Material (Evaluate) The material of the part (steel, aluminum, brass, plastics...). It sets the density and therefore the mass.

Material direction (Features) On which side of the rib line the material goes. It is usually found by itself (toward the part); "Reverse" chooses the other side.

Measure (Evaluate) Measures distances, diameters and angles between selected faces, edges and vertices.

Merge result (Features) Check box of the features that add material. When checked, the material merges with the bodies it touches; when cleared, a new body is created, even if it touches the others.

Merge solids (Features) In the mirror of a body: merges the copy with the original. The body must touch the mirror plane; otherwise the copy stays a separate body, with a warning.

Method E (Drawing) European projection method (first angle, ISO 5456-2): the top view goes below the front view, the left view goes on the right. The symbol in the title block is the truncated cone.

Mid Plane (Options) The feature extends half on one side and half on the other of the sketch plane: the part stays symmetric about the plane.

Midpoint (Relations) The point lies halfway along a line.

Mirror (Features) Copies the chosen features symmetrically about a plane or a flat face.

Mirror Entities (Sketch) Copies the selected entities symmetrically about a centerline and adds the symmetry relations.

Mouse controls (Interface) Wheel: zoom toward the pointer (forward zooms in, backward zooms out), in the drawing too. Ctrl + wheel: rotates the model around the vertical. Dragging: left button in empty space rotates, right button (or Shift + left) pans. On a tablet: two fingers always zoom; to rotate and pan touch the part first, then one finger rotates and two fingers pan. Touch empty space to let it go.

Move Component (Assembly) Moves a component with the mouse or with a finger, only in the degrees of freedom it has left. With the mouse just drag it; the command is for rotating and for the tablet.

Move/Copy Bodies (Features) Rotates a body around an axis through the origin and then moves it. With "Copy the body" the original stays and copies are created, each moved with respect to the previous one.

Multileader (Import) A note with one or more leader lines (MULTILEADER in the DXF or DWG file): arrow, lines, landing and text or block. In the background the arrow is always filled, even if in the file it is of another type; spline leaders are shown as broken lines.

N

New Assembly (Assembly) Another assembly in the same project, with its own components and mates. You switch from one to the other with the list next to the name; right-click its name in the tree to delete it.

New Part (Assembly) Creates an empty part in this project and opens it. The list next to the name, at the top, switches from one part to another.

Normal To (Interface) Turns the view perpendicular to the chosen plane or face, so you see it "straight on". When you enter a sketch the view turns by itself normal to its plane.

Numeric input (Sketch) While you draw you can type the values (length, angle, diameter...) instead of clicking. It also accepts expressions like 40/2. With "Add dimensions with these values" the values become dimensions.

O

OBJ (Import) Text format that describes the surface of the part with points and faces (triangles or polygons). Widely used in 3D graphics; like STL it does not contain the unit of measurement.

OpenCascade (Import) Free library (LGPL license) that reads the solids of STEP and IGES files; FreeCAD uses it too. In Banco CAD it is contained in the page, compressed, and gets ready at the first import (a few seconds): it works without internet too.

Origin (References) The point (0, 0, 0) of the part, where the three standard planes meet. It is best to draw the first sketch starting from the origin: the part stays well placed and symmetries and dimensions become simpler.

Orthographic projections (Drawing) The views of the part obtained by looking at it perpendicularly from the front, from above, from the left... Each view shows two true dimensions.

Over Defined (Sketch) The sketch is red: there are extra dimensions or relations, or ones that contradict each other. The extra dimension or relation must be removed.

Overall dimensions (Drawing) The dimensions of the maximum sizes (length, width, height) plus the diameters of the circles, added automatically.

P

Parallel (Relations) Two or more lines have the same direction.

Parallel mate (Assembly) Two faces, two straight edges or axes, or a face and an edge, parallel to each other. It does not fix the distance.

Parameters (Sketch) The values of the selected entity (coordinates, length, radius...). If the entity is not constrained you can change them by typing.

Perpendicular (Relations) Two lines form a 90° angle.

Perpendicular mate (Assembly) Two faces or two edges at 90°, or an edge perpendicular to a face.

Pitch (Features) How far the helix rises along the axis in one complete turn. In the spiral it is how much the radius grows at each turn. For a spring or a thread the pitch must be larger than the profile, otherwise the coils overlap.

Planes and origin (Assembly) The three standard planes of the part (front, top, right), the planes you have created and its origin, used in a mate. They are chosen in the tree above the view, in the Mate panel or, with "Show them in the view", in the view itself; the plane is drawn when the mouse is over its button. A plane behaves like a face, but it has no "outside": Flip alignment turns it over. The origin behaves like a vertex.

Point (Sketch) Sketch point: it is used as a reference for dimensions or as the position of a hole.

Point on line (Relations) The point lies on the line (or on the circumference), even outside the drawn segment.

Polygon (Sketch) Regular polygon (hexagon, octagon...) inscribed in a construction circle: center and one vertex. The number of sides is chosen in the panel.

Positions (Holes) The sketch points where the centers of the holes go. They are dimensioned as in a normal sketch (for example from the edges of the part).

Preview (Interface) While the PropertyManager is open, the material that will be added is shown in orange and the material that will be removed in red. The part really changes only when you confirm with ✓.

Profile (Features) The closed contour that is extruded, revolved or swept: it is the "section" of the feature.

Profile and Path (Features) The two sketches of the sweep: the profile (closed contour) and the path (lines and tangent arcs) along which the profile moves. Instead of the path sketch you can choose a Helix/Spiral curve.

Profiles (Features) The closed sketches the loft passes through, in the order in which the solid passes through them. They are added with a click in the view or from the "Add" list; the ↑ ↓ arrows change the order, × removes a profile. If the profiles have the same number of sides, each vertex is joined to its corresponding one (the one that twists the solid the least); otherwise the points correspond at the same fraction of the perimeter.

PropertyManager (Interface) The panel that opens on the left when you create or edit a feature: it contains the parameters (depth, radius, number of instances...). ✓ confirms, ✕ cancels.

R

Rebuild (Interface) Recalculating the solid from the tree. It happens by itself every time you change a dimension or a feature (in SolidWorks it is the Rebuild "traffic light").

Reference Plane (References) An extra plane, parallel to a standard plane or to a flat face, at a given distance. It is used to make sketches "in mid-air": for example the end profile of a sweep or a rib halfway along the part.

Reverse Direction (Options) Makes the feature go to the opposite side.

Revolutions (Features) How many turns the helix or the spiral makes around the axis. It need not be a whole number: 5.5 revolutions end on the side opposite to the start.

Revolved Boss/Base (Features) Rotates a closed profile around a centerline: shafts, pins, bushings, knobs. The profile must lie entirely on one side of the axis.

Revolved Cut (Features) Removes material by rotating a profile around an axis: grooves, circular slots, seats for rings.

Rib (Features) Thin reinforcing wall between two parts of the piece, for example in the corner of an angle bracket. You draw only an open line: the material fills the space between the line and the part, with the chosen thickness. The line is extended by itself up to the part.

Right Plane (References) Vertical standard plane containing the Z and Y axes (you look at it from the right). It is perpendicular to the other two.

Rollback bar (Interface) The orange line in the tree. The features below the bar are not calculated: it is used to insert a feature in the middle of the part's history or to see how it was before.

S

Save PDF (Drawing) Creates a PDF file straight away, which opens and prints on any computer, tablet or phone. In the Drawing it saves the vector drawing on the real sheet (A4–A2) at 1:1 scale; in a sketch it saves the vector sketch on a landscape A4 with a standard scale (2:1, 1:1, 1:2...) written at the bottom; otherwise it saves the 3D view as it is on the screen (an image) on a landscape A4. It is in the Project menu and, in the Drawing, among the commands too.

Scale (Drawing) The ratio between the measurements on the sheet and the real ones: 1:2 means half the true size, 2:1 double (standard scales ISO 5455).

Screw in Hole (Assembly) Assembly command: a click on the wall of a hole opens the screws window with the size already chosen (from the thread of the hole, otherwise the largest one that fits), then the piece goes into place by itself with two mates, Concentric on the axis and Coincident on the mouth nearest to the click. The countersunk head stays flush.

Screws and Bolts (Assembly) Standard parts ready to insert into the assembly, from M3 to M20. Socket head ISO 4762, countersunk head ISO 10642, hexagon head ISO 4014 (partial thread) and ISO 4017 (full thread), nut ISO 4032, washer ISO 7089. The sizes are the nominal or maximum ones of the standard, without the chamfers. A screw is a normal part: it can be opened and modified.

Section (Drawing) View obtained by cutting the part with a plane: the cut parts are hatched at 45° (ISO 128-50). The cutting plane is shown with a line and two letters (A-A).

Section View (Evaluate) Cuts the view of the model with a plane to see the inside (it does not modify the part).

Selected Entities (Sketch) The sketch entities you have selected (Ctrl+click to pick more than one). The panel shows the relations that can be added between them.

Self-check (Evaluate) Rebuilds the test parts and compares volume, area, center of mass and overall size with the values calculated by hand. It is used to check that the program calculates correctly (allowed error: 1%).

Shaded isometric view (Drawing) Check box of the drawing: the isometric view is drawn with filled faces and a light, like the "Shaded With Edges" style of SolidWorks. Each face has the color of the part in the 3D view, or the one written in the STEP or IGES file. The orthographic views stay as lines, as the standard requires. The colors are seen on screen, in print and in the SVG; the DXF stays as lines.

Sheet (Drawing) The format of the drawing (A4, A3...) according to ISO 5457, with frame and margins.

Size (Holes) The size of the screw: M6 means metric thread with a nominal diameter of 6 mm.

Sketch (Sketch) 2D drawing on a plane or on a flat face: it is the basis of almost all features. First you draw the shape roughly, then you make it precise with relations and dimensions until the sketch turns black.

Sketch Chamfer (Sketch) Cuts the corner between two lines with an inclined segment (click the vertex).

Sketch Fillet (Sketch) Rounds the corner between two lines with a tangent arc of given radius (click the vertex). Different from the 3D Fillet, which works on the edges of the solid.

Sketch to extrude (Features) The sketch used by the feature. It is chosen from the list or with a click on its lines in the view; "Edit the sketch" reopens it.

Smart Dimension (Sketch) A single command for all dimensions: click a line (length), two lines (distance or angle), a circle (diameter), an arc (radius) or two points; then click where to place the dimension and type the value. It is the SolidWorks dimension.

Snap (Sketch) When you move the pointer close to a notable point (endpoint, midpoint, center, origin) the point "snaps" and the right relation is added, for example Coincident.

Solid Bodies (Features) The separate solids of a part, listed in the "Solid Bodies" folder of the tree. A part can have more than one: they can be kept apart and then combined, subtracted, moved or deleted.

Spline (Import) Free, smooth curve defined by a few points: the control points "pull" it toward themselves (NURBS spline, possibly with weights), or the curve passes exactly through the fit points. It is used for profiles not made of lines and arcs (car bodies, cams, templates). Imported from a DXF it becomes a chain of tangent arcs (biarcs).

Standard values (Holes) The diameters and depths taken from the tables of the standards (ISO 273, DIN 974-1, ISO 15065) for the chosen screw. They can be changed by typing a different value.

Start angle (Features) The point of the circle where the helix starts, as an angle measured from the X axis of the sketch (0° = on the positive X axis). The profile to sweep must be drawn on a plane passing through this point.

Start circle (Features) The sketch with the circle from which the helix or the spiral starts: the diameter of the circle is the start diameter of the curve, the center lies on the axis, the plane of the sketch is perpendicular to the axis. The sketch must contain a single circle.

STEP (Import) Exchange format for 3D models (ISO 10303 standard): it describes solids with their exact surfaces (planes, cylinders, cones, spheres, free-form surfaces) and the unit of measurement. All CAD programs write it; in SolidWorks: File › Save As › STEP AP214. It is the best format to pass a part from one program to another.

STL (Import) Format that describes only the surface of the part, with many triangles. It is the 3D printing format; it exists in a binary version (more compact) and a text version (ASCII). It does not contain the unit of measurement.

Straight Slot (Sketch) Two semicircles joined by tangent lines: it is the shape of the slots used to adjust the position of screws.

Subassembly (Assembly) An assembly used as a component of another one: it moves all together, like a rigid part, and its pieces do not move relative to each other. The bill of materials counts the real parts it contains. To modify it, double-click: its assembly opens.

Suppress (Interface) A suppressed feature stays in the tree (struck through) but is not calculated, as if it were not there. It is useful for trying variants without deleting anything.

Surface area (Evaluate) The sum of the areas of all the faces, in mm²: it is used for example for painting.

Sweep (Features) Moves a closed profile along a path (lines and tangent arcs): bent tubes, handles, wire springs, gaskets. Two sketches are needed: the profile and the path, on different planes; the path starts from the plane of the profile, perpendicular to it. The path can also be a "Helix and Spiral" curve: this is how coil springs and threads are made. In SolidWorks: "Swept Boss/Base".

Swept Cut (Features) Removes material by moving a profile along a path: channels, grooves that turn, seats for tubes.

Symmetric (Relations) Two points (or two entities) are symmetric about a centerline.

T

Tangent (Relations) An arc and a line (or two arcs) touch without a corner, with the same direction at the contact point.

Tangent (Assembly) Mate that makes two surfaces touch without one entering the other: plane and cylinder, plane and sphere, cylinders and spheres with each other. It also works on the free-form surfaces of a STEP file, which have no axis, against a plane or a sphere: there the contact is on the triangles of the surface, so the precision is that of the triangles, not that of the true curve. Flip side chooses on which side they touch, or whether one is inside the other.

Tangent Arc (Sketch) Arc that starts from the end of a line or an arc and continues without a corner (tangent). It is the right way to make sweep paths and slots.

Tangent propagation (Features) Check box of Fillet and Chamfer (checked for new features, as in SolidWorks): with a click on an edge the edges that continue it without a corner are worked too, that is with the same direction at the point where they touch and a face in common. Example: a click on a line of the boundary of a slot fillets all 4 edges (two lines and two arcs). At sharp corners, for example those of a block, the propagation stops.

Taper Helix (Features) Helix whose diameter changes as it rises, like a conical spring: the taper angle is the inclination with respect to the axis. Normally the diameter gets smaller; with "Taper outward" it gets larger.

Tapped (Holes) Hole to be tapped: the model shows the tap drill hole (for M6 Ø5), the thread is shown on the drawing with its designation (M6). ISO coarse pitch.

Thickness (Features) The thickness of the rib, measured perpendicular to the plane of the sketch: half on each side (mid plane) or all on one side.

Thread in section (Assembly) In the section view the thread is drawn with a thin line: at the minor diameter for the screw and at the major diameter for the hole. The hatching goes up to the thick line. Where a screw fills a tapped hole the screw is drawn, covering the hole.

Through All (Options) The cut crosses the whole part, whatever its thickness: if the part changes, the hole stays through.

Title block (Drawing) The box at the bottom right with title, author, date, scale, projection method and general tolerances (ISO 7200).

Top Plane (References) Horizontal standard plane containing the X and Z axes (you look at it from above). It is used for plates, bases and flanges lying flat.

Triad (Interface) The three arrows at the bottom left: X red, Y green, Z blue. They show how the part is oriented in the view. The same colors are used for the dashed axes of the planes.

Trim Entities (Sketch) Removes the part of an entity between two intersections: click the piece to remove. It is used to get closed contours without crossings.

True helix (Assembly) The thread really modeled, with the ISO 68-1 profile turning around the axis, with 32 sides per turn: it is chosen with the check box in the screws window. The part becomes much heavier to calculate and is only useful to see it. On the drawing the thread stays the cosmetic one, as the standard requires.

True surface (Import) Surface written with its formula (plane, cylinder, cone, sphere, torus) instead of as many flat faces: a Ø10 hole stays a cylinder of radius exactly 5. Banco CAD calculates volume and mass on the true surfaces, but on the screen it shows them faceted; in the exported STEP file it writes them true, so the other CAD program measures exact diameters and volumes.

Twist (Features) A blade is twisted when its profile changes inclination from the root to the tip: near the hub it is more inclined, at the tip flatter. It is obtained with a Loft between two or more profiles rotated with respect to each other, on planes at different distances from the axis.

U

Under Defined (Sketch) The sketch (or a part of it) is blue: some dimension or relation is still missing, so something can be moved by dragging it. The part can be built anyway, but it is fragile if you change the measurements.

Undo / Redo (Interface) They go back or forward by one operation (Ctrl+Z and Ctrl+Y). Inside a sketch they undo the single strokes.

User manual (Interface) Project › User manual (PDF) opens the complete manual of the program, to read or print: guided exercise, all the commands and the glossary. It is a separate file, so an Internet connection is needed; it can be downloaded and kept.

V

Vertex (References) The point where three or more edges meet. It is used to measure distances.

Vertical (Relations) The line (or the line joining two points) is parallel to the Y axis of the sketch.

Volume (Evaluate) The space taken up by the solid, in mm^3 ($1\ cm^3 = 1000\ mm^3$).

Z

Zoom to Fit (Interface) Zoom that makes the whole part (or the whole sheet, in the drawing) fit in the window. In SolidWorks it is the F key.

14 Authors and licenses

Banco CAD – Copyright © 2026 Rayuk-creator. The program is free software: you can use, copy and modify it under the **GNU GPL 3** license, with the additional terms of the file CONDIZIONI-AGGIUNTIVE.md. It comes with no warranty. The name “Banco CAD” may only be used to say where the program comes from, not as the name of a modified version.

Parts by other authors included in the program:

Part	What it is for	License
OpenCascade, through occt-import-js	Reading STEP and IGES	LGPL 2.1
LibreDWG, through libredwg-web	Reading DWG files	GPL 3
csg.js	Boolean operations	MIT
IBM Plex	Fonts	OFL 1.1

For the full list: **Project › About and licenses** in the program, or the licenze/ folder of the repository github.com/Rayuk-creator/CAD. SolidWorks, AutoCAD and Inventor are trademarks of their respective owners; Banco CAD is not affiliated with them.

This manual, like the program, is written with the help of Claude. The measurements of the exercise were checked in the program and with hand calculations.